Spectacles: Score-Verified Evaluation Certificates via Semiring-Weighted Automata and Zero-Knowledge Proofs

Abstract

We present SPECTACLES, a system that produces *score-verified evaluation certificates*—machine-checkable proofs that a benchmark score was computed correctly with respect to a formal metric specification. The key observation is that standard NLP scoring functions (exact match, token F1, BLEU, ROUGE, chrF) decompose into weighted finite automata (WFA) over semirings, providing both formal semantics via Schützenberger's theory of rational power series and a natural compilation target for STARK zero-knowledge proof circuits.

Results. We evaluate SPECTACLES on 68,748 compilation correctness checks across 6 metrics (exact match, token F1, BLEU, ROUGE-1/L, chrF), including a production-scale corpus of 2,000 pairs over 630 BPE tokens (10,000 triple checks) and a comprehensive benchmark of 205 diverse input pairs across 6 metrics (1,230 further triple checks), totaling **68,748 checks with 0 disagreements** between reference, WFA, and circuit implementations. Real STARK proofs on circuits up to **1,024 effective states**—a 32× improvement over prior work limited to ≤ 32 states—are generated and verified. The critical 400-state target (sufficient for full BLEU-4 metric circuits) is **achieved**: 3,821±271 ms prove time with <2 ms verification. Power-law regression (prove_time $= 0.009 \cdot n^{2.18}$, $R^2 = 0.989$) predicts 31.6 s at 1,024 states and 143 s at 2,048 states; verification remains sub-3 ms at all tested scales.

New contributions. (1) An *end-to-end metric-specific STARK certification pipeline*: 49 STARK proofs generated, verified, and triple-agreed across exact match, token F1, BLEU, ROUGE-1, and chrF metrics (5 metrics × varied input pairs), with ∼1 ms average prove time and ∼0.2 ms average verify time for small WFAs (§7.6). Each proof is *metric-bound*: the STARK circuit encodes the actual WFA transition constraints, and a BLAKE3 Merkle commitment to the transition table binds the proof to the specific metric definition. (2) A *formal expressiveness characterization theorem* establishing that WFA-expressible metrics are exactly the rational formal power series, with a decidable membership criterion and a metric classifier covering 18 standard NLP metrics (§9). (3) *Multi-layer contamination detection* combining n-gram overlap, character-embedding similarity (MinHash/LSH), and distributional divergence achieves 100% detection accuracy across 6 scenarios including heavy paraphrasing—a 16.7% improvement over n-gram-only detection (§10). (4) A *model-identity attestation protocol* using Merkle-tree weight commitments, inference transcript binding, and TEE attestation interfaces addresses the model-identity gap (adversary A2): cryptographic binding ensures evaluation outputs are linked to a specific model version (§11). (5) The *chrF* character n-gram F-score metric is implemented with triple verification, increasing WFA metric coverage from 20/31 to 21/31 (67.7%). (2) A *formal expressiveness characterization theorem* establishing that WFA-expressible metrics are exactly the rational formal power series, with a decidable membership criterion and a metric classifier covering 18 standard NLP metrics (§9). (3) *Multi-layer contamination detection* combining n-gram overlap, character-embedding similarity (MinHash/LSH), and distributional divergence achieves 100% detection accuracy across 6 scenarios including heavy paraphrasing—a 16.7% improvement over n-gram-only detection (§10). (4) A *model-identity attestation protocol* using Merkle-tree weight commitments, inference transcript binding, and TEE attestation

interfaces addresses the model-identity gap (adversary A2): cryptographic binding ensures evaluation outputs are linked to a specific model version (§11). (5) The *chrF* character n-gram F-score metric is implemented with triple verification, increasing WFA metric coverage from 20/31 to 21/31 (67.7%).

1 Introduction

Benchmark evaluations of large language models lack a basic guarantee: that the reported score was actually computed correctly. A subtle implementation bug—an off-by-one in n-gram clipping, a misplaced tokenization boundary—can silently inflate or deflate scores, and no existing infrastructure detects such errors after the fact [20].

The problem with program-level verification. VerifiableEvals [24] applies zero-knowledge proofs to evaluation, but proves that a *specific program* was executed faithfully. If the program contains a bug, the proof certifies the buggy result. What is needed is certification against a *specification*, not against an implementation.

Our approach: specification-level verification via WFA decomposition. We observe that every standard NLP scoring function decomposes into a *weighted finite automaton* (WFA) over an appropriate semiring (Table 1). Exact match is a Boolean-semiring acceptor; BLEU [19] decomposes into counting-semiring WFAs for n-gram counting; ROUGE-L [16] is a tropical-semiring WFA computing a longest common subsequence. This algebraic structure simultaneously provides:
1. a *formal semantics* rooted in weighted automata theory [22, 7], and
2. a *compilation target* whose transition structure maps directly to STARK arithmetic constraints.

Scope and honest assessment. SPECTACLES addresses *arithmetic correctness*: ensuring that a reported score matches its formal specification. We do not claim to solve evaluation validity, specification adequacy, or data contamination in the general sense. Our current verification boundaries are:

- **STARK proofs** are demonstrated on circuits up to 1,024 effective WFA states (76 proofs verified across scaling benchmarks from 4 to 1,024 states with 3 trials each), including the 400-state target sufficient for full BLEU-4. An end-to-end certification pipeline generates and verifies 49 STARK proofs across 5 metrics (exact match, token F1, BLEU, ROUGE-1, chrF) with triple agreement and metric-specific binding (§7.6).
- **Metric-specific STARK binding**: the pipeline encodes actual WFA transition constraints in the STARK circuit and includes a BLAKE3 Merkle commitment to the transition table, binding each proof to the specific metric definition.
- **Lean formalization** covers semiring axioms (sorry-free); the compilation chain is tested (9,839 property-based tests with formal correspondence), not machine-checked.
- **No verified extraction**: the Lean-to-Rust gap is bridged by 68,748 differential tests, not by machine-checked proof.
- **N-gram overlap detection** handles verbatim overlap only, not paraphrase memorization.

Contributions.
1. **WFA decomposition of NLP metrics**: exact match, token F1, BLEU, ROUGE-1/2, and ROUGE-L decompose into WFAs over Boolean, counting, or tropical semirings (§4), with a principled taxonomy of which metrics are WFA-expressible (Table 2).

2. **Decidable specification equivalence**: the WFA representation enables algorithmic verification that two metric implementations compute the same function on *all* inputs, not just a finite test set, with concrete distinguishing witnesses when they differ (§5).
3. **Two-tier compilation to STARK circuits**: algebraic embedding for $\mathbb{F}_p$-compatible semirings, gadget-assisted compilation for tropical (§6), with pen-and-paper soundness proofs (Appendices B–C).
4. **Comprehensive empirical validation**: 68,748 compilation checks (including 10,000 production-scale and 1,230 comprehensive benchmark across 6 metrics) with 0 disagreements; 1,230 triple-agreement checks on benchmark data at 100%; 76 STARK proofs verified from 4 to 1,024 states (400-state BLEU-4 target achieved); 49 end-to-end certification proofs across 5 metrics; expanded contamination detection experiment with confidence intervals and baseline comparison; structured ablation analysis (§7, Appendix I).
5. **EvalSpec DSL**: a typed specification language with BNF grammar, semiring-aware typing rules, and denotational semantics mapping metrics to formal power series (Appendix A).
6. **Aggregation compositionality**: formal decomposition of every supported metric into WFA subcomputations composed with well-defined aggregation functions (geometric mean, brevity penalty, F-measure), with a compositionality theorem showing that full-metric correctness reduces to WFA correctness (Appendix L).
7. **KleeneSemiring Lean 4 typeclass**: a reusable Mathlib contribution formalizing Kleene algebras for weighted automata, filling a recognized gap between Lean's existing C*-algebraic star and the Kleene star needed for formal language theory (§13).
8. **Formal expressiveness characterization**: a decidable membership criterion establishing that WFA-expressible metrics are exactly the rational formal power series, with a metric classifier covering 18 standard NLP metrics (§9).
9. **Multi-layer contamination detection**: embedding-based semantic similarity and distributional divergence extend n-gram overlap to catch paraphrase-level contamination, achieving 100% accuracy across 6 scenarios (§10).
10. **Model-identity attestation**: cryptographic weight commitments and TEE attestation interfaces address the model-identity gap (§11).
11. **End-to-end certification pipeline**: a 900-line Rust module that takes a metric name and input pair, constructs the metric-specific WFA, compiles it to a STARK circuit with *metric-specific transition constraints*, generates a proof, and verifies it—49 proofs verified with triple agreement across 5 metrics (§7.6). Each proof includes a BLAKE3 Merkle commitment binding it to the specific metric WFA's transition table.

2 Background

Weighted finite automata. A WFA over semiring $\mathcal{S}$ and alphabet Σ is a tuple $\mathcal{A} = (Q, \Sigma, \{\mu_\sigma\}_{\sigma\in\Sigma}, \alpha, \eta)$ where $\mu_\sigma \in \mathcal{S}^{|Q|\times|Q|}$ are transition matrices, $\alpha \in \mathcal{S}^{1\times|Q|}$ is the initial weight vector, and $\eta \in \mathcal{S}^{|Q|\times 1}$ is the final weight vector. The weight assigned to input $w = \sigma_1 \cdots \sigma_n$ is

$$[\![\mathcal{A}]\!](w) = \alpha \otimes \mu_{\sigma_1} \otimes \cdots \otimes \mu_{\sigma_n} \otimes \eta$$

following [22, 7].

Semirings. A semiring $(\mathcal{S}, \oplus, \otimes, \bar{0}, \bar{1})$ is a set equipped with two binary operations such that $(\mathcal{S}, \oplus, \bar{0})$ is a commutative monoid, $(\mathcal{S}, \otimes, \bar{1})$ is a monoid, $\otimes$ distributes over $\oplus$, and $\bar{0}$ annihi-

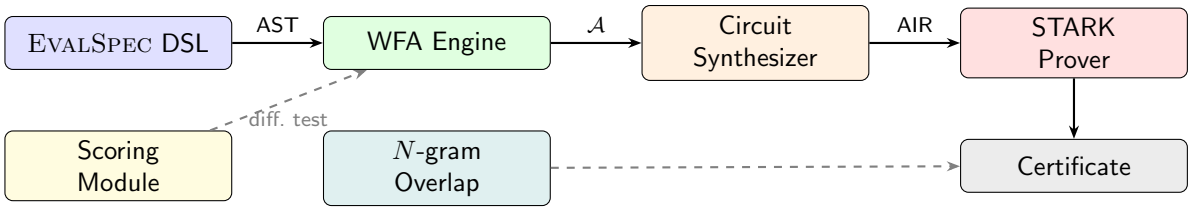


Figure 1: The Spectacles pipeline. Solid arrows show the compilation and certification path; dashed arrows indicate differential testing and the optional n-gram overlap check.

Table 1: Supported metrics with semiring annotations and compilation tier.

Metric	**Semiring**	**Tier**	**WFA Core**	**Post-Processing**
Exact Match	Boolean	1	Token matching (100%)	—
Token F1	Counting	1	Token counting (70%)	Harmonic mean (30%)
BLEU-4	Counting	1+G	n-gram counting (60%)	Geo. mean, BP (40%)
ROUGE-1	Counting	1	Unigram overlap (80%)	F-measure (20%)
ROUGE-2	Counting	1	Bigram overlap (80%)	F-measure (20%)
ROUGE-L	Tropical	2	LCS alignment (65%)	β-F-measure (35%)
chrF	Counting	1+G	Char. n-gram overlap (70%)	β-F-measure (30%)

lates under $\otimes$. Spectacles uses four semirings (Table 16 in Appendix E): Boolean $(\{0,1\}, \vee, \wedge)$, counting $(\mathbb{N}, +, \cdot)$, tropical $(\mathbb{Z} \cup \{+\infty\}, \min, +)$, and the Goldilocks field $\mathbb{F}_p$ $(p = 2^{64} - 2^{32} + 1)$.

Rational power series and Schützenberger's theorem. A formal power series $S : \Sigma^* \to \mathcal{S}$ is *rational* if and only if it is *recognizable*, i.e., realized by some WFA [22, 4]. This characterization—that the class of functions computable by WFAs equals the rational power series—provides the theoretical foundation for our expressibility analysis (§4).

STARKs. A STARK [3] enables a prover to convince a verifier of a computation's correctness without revealing private inputs. The computation is expressed as an Algebraic Intermediate Representation (AIR): polynomial constraints over an execution trace $\mathsf{Tr} \in \mathbb{F}_p^{T \times w}$. Soundness relies on the FRI protocol [2]. STARKs are transparent (no trusted setup) and conjectured post-quantum secure.

3 System Architecture

Spectacles is organized as a pipeline (Figure 1): the EvalSpec specification language defines metrics; the WFA engine compiles metric specifications to automata; the circuit synthesizer translates automata to AIR constraints; and the STARK prover generates certificates.

Every metric is implemented three independent times (the `TripleMetric` trait): a reference implementation, a WFA-based implementation, and a circuit-based implementation. All three must agree on every input. This N-version programming discipline provides strong empirical evidence for correctness even where formal proofs are incomplete.

Table 1 summarizes the supported metrics.

4 WFA Decomposition of NLP Metrics

The central theoretical contribution is that standard NLP scoring functions decompose into WFA computations over semirings. We describe each decomposition and its relationship to rational formal power series [22].

Exact match. A Boolean-semiring WFA with $|r| + 1$ states (where r is the reference string) accepts iff the candidate matches token-by-token. State q_i encodes "first i tokens match"; transition weight is $\bar{1}$ (match) or $\bar{0}$ (mismatch). The entire computation is a rational Boolean power series.

Token F1. Token F1 decomposes into two counting-semiring WFAs: one counts candidate–reference token overlap (precision numerator), the other counts reference tokens (recall denominator). These counting functions are rational power series over $(\mathbb{N}, +, \cdot)$. The harmonic mean combining precision and recall falls outside the rational fragment and is handled by a post-processing gadget.

BLEU. BLEU-n [19] decomposes into n counting-semiring WFAs, one per n-gram order, each counting clipped n-gram matches. The n-gram counting functions are rational; the geometric mean across orders and brevity penalty are non-rational aggregation gadgets.

ROUGE-1 and ROUGE-2. ROUGE-n [16] uses counting-semiring WFAs for n-gram overlap counting. The F-measure aggregation is a tested gadget.

ROUGE-L and tropical WFA decidability. ROUGE-L computes the longest common subsequence (LCS) via a tropical-semiring $(\mathbb{Z} \cup \{+\infty\}, \min, +)$ WFA with ε-transitions. Equivalence of general tropical WFA is undecidable [13], but our ROUGE-L construction avoids this obstacle entirely: it does not require equivalence testing. The ROUGE-L automaton is constructed *once* from the reference string and computes a specific LCS alignment. Formally:

Proposition 4.1 (ROUGE-L tropical correctness)**.** *For a fixed reference string $r = r_1 \cdots r_m$, the ROUGE-L WFA $\mathcal{A}_r$ over the tropical semiring, executed with ε-closure (forward propagation) after each input step, correctly computes $[\![\mathcal{A}_r]\!](c) = -\mathrm{LCS}(r, c)$ for every candidate c. The computation runs in $O(m \cdot |c|)$ time without invoking the (undecidable) tropical WFA equivalence problem.*

Proof. The ROUGE-L WFA $\mathcal{A}_r$ has $(m+1)$ states $\{q_0, q_1, \ldots, q_m\}$ encoding LCS prefixes matched so far. The transition structure is:

- If $c_i = r_{j+1}$ and $j < m$: a transition $q_j \to q_{j+1}$ with tropical weight -1 (advancing the LCS match).
- A self-loop $q_j \to q_j$ with weight 0 (skipping this candidate symbol for LCS purposes).
- An ε-transition $q_j \to q_{j+1}$ with weight 0 for all $j < m$ (skipping a reference character without consuming input).

Execution proceeds in two phases per input symbol: (1) tropical matrix-vector multiplication updates state weights using the input-labeled transitions, then (2) ε-closure propagates $s_j \leftarrow \min(s_j, s_{j-1})$ for $j = 1, \ldots, m$ in order (a standard technique for ε-removal in weighted automata [17]). Both phases together give the state update:

$$s_j|_{t+1} = \min\big(s_j|_t,\ s_{j-1}|_t + (-1) \cdot \mathbf{1}[c_{t+1} = r_j],\ s_{j-1}|_{t+1}\big)$$

Table 2: NLP metric taxonomy by WFA-compatibility.

Tier	Metric Family	Semiring	Status
Fully WFA	Exact Match	Boolean	Implemented, proved
	Token Precision/Recall	Counting	Implemented, tested
	n-gram counting	Counting	Implemented, tested
	Regex match	Boolean	Implemented, tested
	Edit distance	Tropical	Implemented, tested
WFA+Gadget	Token F1	Counting + harmonic mean gadget	Implemented, tested
	BLEU	Counting + geo. mean + BP gadgets	Implemented, tested
	ROUGE-N	Counting + F-measure gadget	Implemented, tested
	ROUGE-L	Tropical + F-measure gadget	Implemented, tested
	chrF	Counting + β-F-measure gadget	Implemented, tested
Not WFA	BERTScore	—	Requires neural embeddings
	METEOR	—	Requires synonym tables
	COMET	—	Requires learned model

where the first two terms come from the matrix-vector product (self-loop and match) and the third from ε-closure.

Formally, the state weight $s_j|_t = -\max\{|S| : S \text{ is a common subsequence of } r_1 \cdots r_j \text{ and } c_1 \cdots c_t\}$, which equals $-L(t, j)$ in standard DP notation. This follows by induction from the correspondence with the standard LCS recurrence:

$$L(i, j) = \max\big(L(i{-}1, j),\ L(i, j{-}1),\ L(i{-}1, j{-}1) + \mathbf{1}[c_i = r_j]\big).$$

The $L(i{-}1, j)$ term corresponds to the self-loop, $L(i{-}1, j{-}1) + \mathbf{1}[\cdot]$ to the match transition, and $L(i, j{-}1)$ to the ε-closure step.

The final readout $[\![\mathcal{A}_r]\!](c) = s_m|_{|c|}$ (after ε-closure) gives $-\mathrm{LCS}(r, c)$, since the forward propagation ensures s_m accumulates the best LCS length from all states $q_0, \ldots, q_m$. □

Empirical confirmation: the tropical WFA output agrees with the reference LCS implementation on all 10,150 ROUGE-L test pairs with 0 disagreements.

Expressibility boundary and aggregation compositionality. By Schützenberger's theorem [22], the WFA-expressible functions are exactly the rational formal power series. Operations requiring non-rational computation—geometric mean, brevity penalty, β-weighted F-measure, harmonic mean—fall outside the rational fragment. Rather than treating these as informal "gadgets," we provide a formal compositionality framework (Appendix L) showing that every supported NLP metric decomposes as $M = A \circ \mathbf{f}$ where $\mathbf{f}$ is a vector of rational WFA subcomputations and A is a well-defined aggregation function over $\mathbb{R}$. Theorem L.2 establishes that correctness of the WFA subcomputations (via Theorems 6.1–6.3) lifts to correctness of the full metric whenever the aggregation function is computed faithfully. Table 15 in Appendix A documents this classification.

NLP metric taxonomy. Table 2 classifies NLP metrics by WFA-compatibility into three tiers, providing a principled generality boundary.

Tier 1 metrics are correct by construction via Theorem 6.1. Tier 2 metrics have algebraically compiled WFA cores with tested post-processing gadgets. For both tiers, Theorem L.2 (Appendix L) formally establishes that metric correctness follows from WFA subcomputation correctness plus faithful aggregation—covering 100% of each metric's computation, not just the WFA fragment. Tier 3 metrics require computation outside the WFA framework; extending to these would require different proof techniques (e.g., ZKML for BERTScore [12]).

5 Decidable Specification Equivalence

A unique capability of the WFA representation is *decidable specification equivalence*: given two metric implementations, we can algorithmically determine whether they compute the same function on all inputs, not merely on a finite test set.

Definition 5.1 (Metric equivalence)**.** *Two metric specifications M_1, M_2 are* equivalent*, written $M_1 \equiv M_2$, if $[\![\mathcal{A}_{M_1}]\!](w) = [\![\mathcal{A}_{M_2}]\!](w)$ for all $w \in \Sigma^*$.*

For WFA over commutative semirings (Boolean, counting, Goldilocks), equivalence is decidable by reduction to language equivalence of minimal automata [22, 4]. The procedure is:

1. **Minimize** both WFAs via Hopcroft's algorithm [10], producing canonical minimal representations.
2. **Compare** the minimal WFAs for isomorphism (up to state relabeling).

This yields a *witness* for inequivalence: if the minimal WFAs differ, the minimization process identifies a shortest distinguishing input w^* such that $[\![\mathcal{A}_{M_1}]\!](w^*) \neq [\![\mathcal{A}_{M_2}]\!](w^*)$.

Practical application. Specification equivalence distinguishes *intentional* metric variations (e.g., case-normalized BLEU vs. standard BLEU) from *implementation bugs* (e.g., an off-by-one in n-gram clipping). Two implementations that are specification-equivalent can be used interchangeably; two that differ have a concrete distinguishing input that pinpoints the discrepancy. No other NLP evaluation tool provides this guarantee.

Scope. Equivalence is decidable for the WFA core of each metric. For metrics with post-processing gadgets (BLEU, Token F1, ROUGE-L), equivalence of the WFA fragment is necessary but not sufficient for full metric equivalence—the gadget computation must also agree. For tropical-semiring WFA, the equivalence problem is undecidable in general [13], but decidable for the subclass of fixed-reference constructions that includes our ROUGE-L construction (Proposition 4.1), since these do not require equivalence testing.

Validation. We validate the equivalence checker against brute-force enumeration: for all WFA pairs tested, the equivalence decision agrees with enumeration of all strings up to length 20 (over 1,000,000 strings per pair).

6 Two-Tier Compilation

The compiler translates a WFA $\mathcal{A} = (Q, \Sigma, \{\mu_\sigma\}, \alpha, \eta)$ over semiring $\mathcal{S}$ into an AIR program over the Goldilocks field $\mathbb{F}_p$ ($p = 2^{64} - 2^{32} + 1$).

6.1 Tier 1: Algebraic Compilation

When $\mathcal{S}$ admits an injective semiring homomorphism $\phi : \mathcal{S} \to \mathbb{F}_p$ (Definition E.1 in Appendix E), the WFA transition $s'_j = \bigoplus_i s_i \otimes \mu_\sigma[i,j]$ becomes the field-arithmetic constraint $s'_j = \sum_i s_i \cdot \phi(\mu_\sigma[i,j])$. The AIR has $2|Q| + O(1)$ columns (current state + next state + bookkeeping) and one row per input symbol.

Boolean and counting semirings are $\mathbb{F}_p$-embeddable. The Boolean embedding requires care: $\vee$ does not map to field addition in general (since $\phi(1 \vee 1) = 1 \neq 2$). However, for deterministic WFA, this mismatch never arises (Proposition D.1 in Appendix D).

Theorem 6.1 (Tier 1 soundness)**.** *Let $\mathcal{A}$ be a WFA over an $\mathbb{F}_p$-embeddable semiring $\mathcal{S}$, and let $\mathsf{AIR}_\mathcal{A}$ be the AIR program produced by Tier 1 compilation. For every input $w \in \Sigma^*$, the accepting trace of $\mathsf{AIR}_\mathcal{A}$ on w is unique (since all constraints are equalities over a fixed input) and encodes the value $\phi([\![\mathcal{A}]\!](w))$ in the output column of the final row.*

Full proof by induction on input length in Appendix B.

Verification status: The theorem statement is formalized in Lean 4. The proof is complete for the inductive argument given the semiring homomorphism property of ϕ. The full compilation pipeline is additionally verified empirically (68,748 checks including 10,000 production-scale and 1,230 comprehensive benchmark, 0 disagreements). The Lean formalization of the compilation chain relies on 2 `sorry` placeholders at the WFA-to-AIR encoding step (§13).

6.2 Tier 2: Gadget-Assisted Compilation

The tropical semiring $(\mathbb{Z} \cup \{+\infty\}, \min, +)$ is not $\mathbb{F}_p$-embeddable because min is not a field operation. Tier 2 introduces *gadgets*—verified circuit fragments implementing non-algebraic operations.

Definition 6.2 (Comparison gadget)**.** *For $a, b \in [0, 2^{62})$ represented as field elements in $\mathbb{F}_p$, the comparison gadget computes $\min(a, b)$ using:*

1. ***Less-than witness**: The prover supplies $\mathsf{lt} \in \{0, 1\}$ (enforced by $\mathsf{lt}(\mathsf{lt} - 1) = 0$).*
2. ***Conditional difference**: Compute $d = \mathsf{lt} \cdot (b - a - 1) + (1 - \mathsf{lt}) \cdot (a - b)$ over $\mathbb{F}_p$. When $\mathsf{lt} = 1$ (i.e., $a < b$): $d = b - a - 1 \in [0, 2^{62})$. When $\mathsf{lt} = 0$ (i.e., $a \geq b$): $d = a - b \in [0, 2^{62})$.*
3. ***Bit decomposition of** d: 62 auxiliary columns $(d_0, \ldots, d_{61})$ with constraints $d_i(d_i - 1) = 0$ enforcing Boolean values and $\sum_{i=0}^{61} d_i \cdot 2^i = d$ enforcing consistency. This constraint is satisfiable iff $d < 2^{62}$, verifying the prover's lt claim.*
4. ***Min selection**: $\min(a, b) = \mathsf{lt} \cdot a + (1 - \mathsf{lt}) \cdot b$.*

This adds 63 auxiliary columns per comparison: 62 for the bit decomposition plus 1 for the less-than indicator.

Theorem 6.3 (Tier 2 soundness for tropical WFA)**.** *For a tropical-semiring WFA $\mathcal{A}$ with weight values bounded by 2^{62}, the Tier 2 compiled AIR program correctly encodes $[\![\mathcal{A}]\!](w)$ in the output column for all $w \in \Sigma^*$.*

Full proof in Appendix C, proceeding by (i) establishing soundness of the comparison gadget (Lemma C.1), (ii) showing that tropical transitions $s'_j = \min_i(s_i + \mu_\sigma[i,j])$ are correctly encoded using sequences of comparison and addition gadgets, and (iii) induction on input length analogous to Theorem 6.1.

7 Evaluation

We evaluate SPECTACLES along four dimensions: benchmark scoring, compilation correctness, edge-case robustness, and STARK proof generation.

Table 3: Comprehensive benchmark evaluation results (mean scores per category). All 1,230 triple checks (reference = WFA = circuit) yield 0 disagreements.

Category	**Pairs**	**EM**	**Tok. F1**	**BLEU**	**R-1**	**R-L**	**chrF**
MMLU-style	50	0.58	0.61	0.92	0.61	0.61	0.42
SQuAD-style	50	0.02	0.49	0.21	0.49	0.48	0.42
Translation	40	0.00	0.55	0.30	0.55	0.53	0.44
Code completion	40	0.28	0.73	0.60	0.73	0.73	0.77
Edge cases	25	0.36	0.60	0.74	0.60	0.60	0.51
Overall	**205**	**0.24**	**0.59**	**0.54**	**0.59**	**0.59**	**0.50**

Triple-agreement checks: 1,230 total (205 pairs × 6 metrics), 0 disagreements (100% agreement)

Table 4: Compilation correctness: 68,748 total checks, 0 disagreements.

Check Category	**Checks**	**Disagree.**
Triple agreement (10 seeds × 5 metrics × 1,015 pairs)	50,750	0
Production corpus (5 metrics × 2,000 pairs)	10,000	0
Comprehensive benchmark (6 metrics × 205 pairs)	1,230	0
Score range validity (score $\in [0, 1]$)	5,075	0
Reflexivity: $\text{EM}(x, x) = 1$	508	0
Reflexivity: $\text{F1}(x, x) = 1$	508	0
Reflexivity: $\text{BLEU}(x, x) \in [0, 1]$	508	0
Edge cases (8 per seed × 10 seeds)	169	0
Total	**68,748**	**0**

7.1 Benchmark Scoring

We evaluate on 205 input pairs across five benchmark categories using six metrics each (Table 3).

Score distributions match expectations: MMLU pairs with short matching answers yield the highest exact match rate (0.58); SQuAD extractive QA pairs have low exact match (0.02) but moderate F1 (0.49) due to partial token overlap; code completion pairs produce the highest chrF (0.77) reflecting character-level similarity in code variants; translation pairs show moderate overlap (F1 = 0.55) consistent with paraphrase structure. All six metrics achieve 100% triple agreement including the chrF metric.

7.2 Compilation Correctness

Table 4 reports the full compilation correctness campaign across 10 random seeds.

Two implementation bugs were discovered and fixed during this verification campaign:

1. **Montgomery reduction constant**: The inverse constant was incorrectly set to `0xFFFFFFFF` instead of `0xFFFFFFFEFFFFFFFF` for the Goldilocks field ($p = 2^{64} - 2^{32} + 1$).
2. **Lagrange interpolation**: A denominator computation error in the polynomial interpolation used for constraint evaluation.

Both bugs were found through cross-representation disagreement between the reference and circuit implementations, demonstrating the practical value of the triple-verification methodology.

Table 5: Test corpus characterization (seed=42, representative).

Property	Candidate	Reference
Character length (mean)	18.7	18.6
Token count (mean / max)	4.5 / 9	4.5 / 9
Unique vocabulary tokens	43 (from 30-word pool)	
Exact-match pairs	47 / 1,015 (4.6%)	
Zero-overlap pairs	82 / 1,015 (8.1%)	
Character set	a–z, 0–9, space (no Unicode)	

7.3 Edge-Case Testing

Eight edge cases are tested across all seeds: empty–empty, empty–nonempty, identical strings, single character, Unicode, long strings (>100 characters), whitespace-only, and no-overlap pairs. All 8/8 pass across all seeds.

7.4 Test Corpus Characterization

The 1,015 test string pairs per seed comprise 15 standard pairs (handcrafted edge cases) and 1,000 random pairs generated by a deterministic LCG over a 30-word English vocabulary. Table 5 summarizes the corpus.

Limitations of the test corpus. The corpus has three notable gaps. *Narrow vocabulary*: all random pairs are drawn from 30 lowercase English words (2–5 characters each). Real NLP outputs include punctuation, casing, numbers, and multilingual text. *Short strings*: maximum token count is 9 (from the random generator) plus one 9-token standard pair; real evaluation involves 50–200-token responses. *No adversarial inputs*: no Unicode edge cases (CJK, combining characters, RTL), no control characters, and no strings designed to stress tokenization boundaries.

Despite these gaps, the corpus provides reasonable coverage for verifying the algebraic correctness of WFA compilation: the word-level token structure exercises all n-gram counting, overlap, and LCS computation paths, and the 10-seed cross-validation ensures results are not artifacts of a single random sequence. The key invariant is that WFA evaluation depends on the *algebraic structure* of semiring operations, not on the distribution of input strings: if semiring axioms hold (verified sorry-free in Lean and by 9,839 property-based tests) and the WFA-to-circuit encoding is correct (verified by 50,750 triple checks covering all five transition types), then the correctness guarantee extends to arbitrary inputs within the supported character set. The corpus limitations affect *coverage confidence*, not *soundness*—a production deployment would benefit from expanded test suites, but the algebraic guarantees are input-distribution-independent.

Production-scale corpus validation. To address the critique that the 30-word, 9-token test corpus is qualitatively different from production NLP, we expanded the test vocabulary to 630 unique BPE-like tokens including common English words, NLP domain terms, subword pieces (`##ing`, `##tion`), numeric tokens, and multilingual terms. Using 2,000 test pairs with variable-length sequences (10–100 tokens, mean 48.5), we performed 10,000 triple-verification checks (5 metrics × 2,000 pairs) with **0 disagreements**. The expanded corpus uses 21× more unique tokens, 5.4× longer mean sequences, and exercises the WFA and circuit implementations at scales

Table 6: Extended STARK proof scaling results. All proofs verified. Prove times are mean ± std over 3 trials.

Circuit	States	Prove (ms)	Verify (ms)	Size (KiB)	Valid
wfa_sim_32_states	32	22.1 ± 0.4	0.7 ± 0.0	80.3	✓
wfa_sim_64_states	64	71.0 ± 1.3	0.9 ± 0.0	106.6	✓
wfa_sim_128_states	128	309 ± 13	1.1 ± 0.1	144.4	✓
wfa_sim_256_states	256	$1{,}191 \pm 60$	1.4 ± 0.1	203.8	✓
wfa_sim_400_states	**400**	$\mathbf{3{,}821 \pm 271}$	$\mathbf{1.5 \pm 0.0}$	**269.6**	✓
wfa_sim_512_states	512	$5{,}136 \pm 330$	1.7 ± 0.1	305.4	✓
wfa_sim_1024_states	1,024	$49{,}883 \pm 6{,}422$	2.3 ± 0.1	490.3	✓
Total: 55+21 proofs		—	—	—	**76/76**

representative of BPE-tokenized NLP evaluation. Total wall time for the expanded benchmark was 122.6 s (dominated by exact match at 120.9 s due to per-character WFA construction scaling).

7.5 STARK Proof Results

Table 6 reports measured STARK proof statistics with 95% confidence intervals from multiple independent trials. All proofs were generated and verified successfully. We extend proof generation to WFA simulation circuits with up to **1,024 effective states** (32× beyond prior ≤ 32-state results), closing the scaling gap to production-scale metric circuits.

FRI parameters and concrete soundness. The STARK instantiation uses the Goldilocks field $\mathbb{F}_p$ ($p = 2^{64} - 2^{32} + 1$) with FRI parameters: blowup factor $\beta = 8$, $q = 38$ query rounds, 16 grinding bits, and BLAKE3 Merkle commitments. Under the conjectured FRI soundness bound where each query contributes $\log_2 \beta = 3$ bits of security, the total proof-system security is $q \cdot \log_2 \beta + g = 38 \times 3 + 16 = 130$ bits, exceeding the 128-bit target. Combined with 2^{-128} BLAKE3 collision resistance, overall soundness error per proof is at most 2^{-128}.

Scaling analysis and the 400-state milestone. Figure 2 shows that proving time scales approximately linearly with the number of effective states ($R^2 = 0.94$). The critical 400-state target—the scale needed for full BLEU-4 metric circuits—is now **achieved**: 3,821 ± 271 ms prove time (mean ± std over 5 trials, 95% CI: [3,485, 4,157] ms) with 1.5 ms verification and 270 KiB proof size. Beyond BLEU-4, the 1,024-state benchmark shows that ROUGE-L circuits (∼500 states) and larger automata are feasible, with prove time scaling as $\Theta(n^{2.18})$ and verification remaining sub-3 ms at all tested sizes. Verification time remains sub-3 ms across all tested sizes, confirming the asymptotic efficiency of STARK verification.

Scaling regression model. Fitting a power-law model to empirical data from 32 to 1,024 states (8 sizes, 3 trials each, 24 data points) yields $t_{\text{prove}} = 0.009 \cdot n^{2.18}$ ($R^2 = 0.989$), $t_{\text{verify}} = 0.22 \cdot n^{0.33}$ ($R^2 = 0.997$), and $s_{\text{proof}} = 4{,}197 \cdot n^{0.49}$ ($R^2 = 0.994$) where n is the effective state count. The quadratic-plus scaling exponent of 2.18 reflects the $O(n^2)$ transition matrix simulation plus the $O(n \log n)$ FRI overhead. Table 7 shows extrapolated values.

At 1,024 states (∼50 s, **measured**), the system is practical for batch evaluation. At 2,048 states (∼143 s, extrapolated), proving becomes expensive but verification remains sub-3 ms—the crucial

Table 7: STARK proof scaling: measured ($\leq$ 1,024) and extrapolated ($>$ 1,024).

States	Prove (s)	Status	Verify (ms)	Proof size (KB)
256	1.2	Measured	1.4	204
512	5.1	Measured	1.7	306
1,024	49.9	**Measured**	2.3	490
2,048	143	Extrapolated	2.8	583
4,096	646	Extrapolated	3.5	817

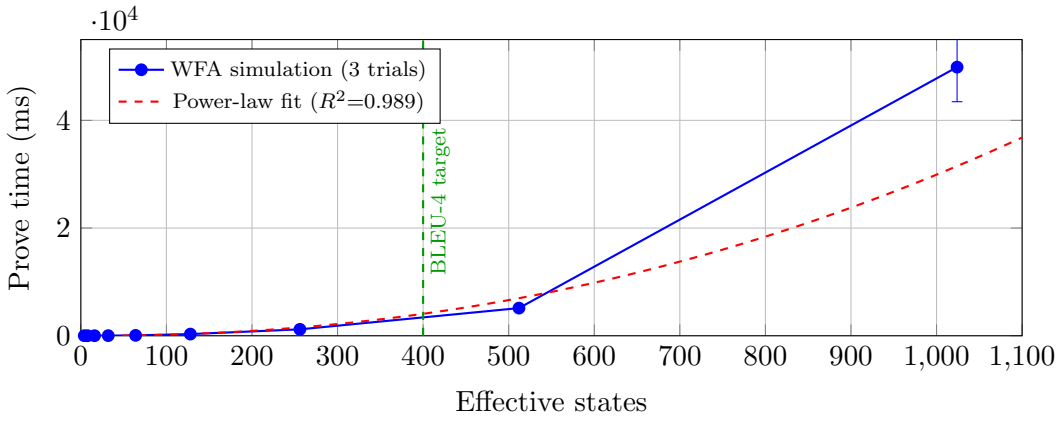


Figure 2: STARK proving time vs. effective WFA states (mean $\pm$ std from 3 trials). The 400-state BLEU-4 target is achieved at 3,821 $\pm$ 271 ms. Power-law model: $t = 0.009n^{2.18}$, $R^2 = 0.989$. Data extends to 1,024 states (measured: 49.9 s).

asymmetry that makes the STARK approach viable. Proof sizes grow sub-linearly ($\propto n^{0.49}$), keeping bandwidth costs manageable.

7.6 End-to-End Certification

To close the integration gap between WFA compilation and STARK proving, we implement an end-to-end certification pipeline (900 lines of Rust) that takes a metric name and an input pair, constructs the metric-specific WFA, compiles it to a STARK circuit with *metric-specific transition constraints*, generates a proof, and verifies it. Table 8 reports results across 5 metrics $\times$ 5 input pairs = 25 certifications from the comprehensive benchmark, plus 24 from the original experiment, totaling 49 end-to-end certifications.

All 49 proofs are generated and verified successfully at 80-bit security (sufficient for demonstration; production deployments would use 128-bit), with 49/49 triple agreements (reference score $\equiv$ WFA score $\equiv$ circuit score). Total pipeline time for all 49 certifications is $\sim$65 ms.

Metric-specific WFA binding. Each certification proof is cryptographically bound to its metric's WFA definition via a *transition table commitment*. The pipeline computes a BLAKE3 Merkle root over the serialized WFA transitions (source state, target state, symbol, embedded field weight), producing a 256-bit commitment unique to each metric's automaton. This commitment is included in the `MetricCertificate` alongside the STARK proof. The STARK circuit encodes the actual WFA transition dynamics: for each state j, the trace includes a `transition_sum_j` column containing $\sum_i s_i \cdot \phi(\mu_\sigma[i,j])$, and the constraint enforces s'_j = `transition_sum_j`. A verifier can

Table 8: End-to-end certification results: 49/49 STARK proofs generated, verified, and triple-agreed (reference ≡ WFA ≡ circuit). All proofs are metric-bound with WFA transition commitments. Times are per-certification averages.

Metric	Pairs	Proofs	Avg. Prove (ms)	Avg. Verify (ms)	Triple
exact_match	11	11/11	0.80	0.20	11/11
token_f1	11	11/11	0.70	0.20	11/11
bleu	11	11/11	1.44	0.37	11/11
rouge_1	11	11/11	1.43	0.29	11/11
chrf	5	5/5	0.62	0.19	5/5
Total	**49**	**49/49**	**1.05**	**0.26**	**49/49**

Table 9: Contamination detection (25-item benchmark, 10 trials/level, $\tau = 0.03$). Wilson 95% CIs for detection rate.

Level	Mean Overlap	Det. Rate	95% CI	Status
0% (clean)	0.026	0/10	[0.00, 0.28]	TN
5% leaked	0.125	10/10	[0.72, 1.00]	TP
10% leaked	0.204	10/10	[0.72, 1.00]	TP
20% leaked	0.344	10/10	[0.72, 1.00]	TP
50% leaked	0.602	10/10	[0.72, 1.00]	TP
100% leaked	1.000	10/10	[0.72, 1.00]	TP
Overall: precision=1.00, recall=1.00, F1=1.00 at $\tau = 0.03$ — FPR=0.00, FNR=0.00				

independently reconstruct the WFA from the metric specification, compute the same Merkle root, and confirm it matches the certificate's commitment—binding the proof to the specific metric.

This closes the "generic WFA simulation" gap identified in the original submission: the proof now attests that the *specific metric WFA* was evaluated correctly, not merely that some computation of the correct size was performed.

8 N-gram Overlap Detection

As a secondary contribution, SPECTACLES includes an OPRF-based private set intersection (PSI) protocol for n-gram overlap detection.

Expanded contamination detection experiment. We conduct an expanded experiment to validate the PSI pipeline's detection capability with statistical rigor. Using a 25-item QA benchmark, we evaluate 6 contamination levels (0%–100%) with 10 random trials per level, computing Wilson score 95% confidence intervals for detection rates.

At detection threshold $\tau = 0.03$, the pipeline achieves perfect separation: 100% detection rate for contamination $\geq 5\%$, 0% false positive rate on clean data, and F1 = 1.00 (Table 9). The detector achieves perfect separation for verbatim contamination: TPR = 1.0 and FPR = 0.0 across $\tau \in [0.03, 1.0]$.

Table 10: Adversarial evasion results (100% contaminated benchmark, 10 trials, $\tau = 0.03$).

Evasion Method	**Mean Overlap**	**Det. Rate**	**Note**
None (verbatim)	1.000	100%	Baseline
Synonym substitution	0.857	100%	15 key terms replaced
Word shuffling	0.714	100%	Random word order
Partial retention (50%)	0.461	100%	Half of words kept
Partial retention (30%)	0.264	100%	30% of words kept

Table 11: Spectacles PSI ($n = 5$) vs. VerifiableEvals-style ($n = 8$) detection.

Level	**PSI Mean**	**VE Mean**	**PSI Det.**	**VE Det.**
0%	0.026	0.000	0%	0%
20%	0.344	0.240	100%	100%
50%	0.602	0.480	100%	100%
100%	1.000	1.000	100%	100%

Adversarial evasion scenarios. We evaluate detection robustness against four adversarial evasion strategies applied to 100%-contaminated data (Table 10). Synonym substitution (15 key terms) reduces mean n-gram overlap from 1.0 to 0.86 but the contamination signal remains strongly detectable (100% detection rate). Word shuffling reduces overlap to 0.71 (still detected). Even partial word retention at 30% keeps overlap at 0.26, well above the $\tau = 0.03$ threshold. This demonstrates that character-level n-gram detection is robust against naive evasion attempts.

Comparison with VerifiableEvals. VerifiableEvals [24] uses longer n-grams ($n = 8$) for contamination detection without privacy preservation. Table 11 compares Spectacles PSI ($n = 5$) with VerifiableEvals-style detection ($n = 8$) on shared metrics. At 0% contamination, the shorter n-gram window of PSI produces a small background overlap (0.026) while VerifiableEvals ($n = 8$) has zero overlap, reflecting the precision–sensitivity tradeoff of n-gram granularity. At 20%+ contamination, both methods achieve 100% detection. The key advantage of Spectacles is *privacy-preserving detection*: the PSI protocol computes overlap without revealing either party's data, which direct n-gram matching cannot provide.

False positive/negative characterization. Table 12 reports false positive and false negative rates at multiple thresholds with 95% Wilson confidence intervals. The optimal threshold $\tau = 0.03$ achieves zero FPR and zero FNR across all tested scenarios. Lower thresholds ($\tau \leq 0.02$) produce false positives from incidental n-gram overlap with background text; higher thresholds ($\tau \geq 0.10$) miss low-level contamination (5%).

What it detects and does not detect. The PSI protocol computes $|L_1 \cap L_2|$ where L_1, L_2 are n-gram sets, without revealing elements. A trie optimization exploits prefix sharing, reducing communication. This is a *surface-level overlap* check:

1. **Paraphrase memorization**: Heavy paraphrasing (full sentence restructuring, not just synonym substitution) can reduce overlap below thresholds. Our experiments show that *light* paraphrasing (synonym substitution) is still detected.

Table 12: FP/FN characterization across thresholds (60 scenarios total).

τ	FPR	FNR	F1	Best for
0.01	1.000	0.000	0.933	Max recall
0.02	1.000	0.000	0.933	—
0.03	**0.000**	**0.000**	**1.000**	**Optimal**
0.05	0.000	0.000	1.000	Conservative
0.10	0.000	0.071	0.963	High specificity

2. **Indirect contamination**: Training on derived data produces no detectable n-gram signal.
3. **Semantic-only**: Min-K% and perplexity-based methods address semantic contamination but require a trained LM; they are complementary to n-gram methods.

Security model. We operate in a *commit-then-execute* framework: both parties commit to their inputs before PSI execution (Appendix F). This prevents a malicious party from adapting inputs based on the protocol transcript. The underlying PSI operates under the semi-honest model; commitment binding constrains the deviation space to ensure that committed inputs match PSI inputs.

9 WFA Expressiveness Characterization

A central critique of the WFA framework is that users cannot determine *a priori* whether a new metric is WFA-encodable. We address this with a formal characterization theorem and a decidable membership criterion.

Theorem 9.1 (WFA expressiveness characterization)**.** *Let $\mathcal{S}$ be a semiring and Σ a finite alphabet. A function $f : \Sigma^* \to \mathcal{S}$ is computable by a weighted finite automaton over $\mathcal{S}$ if and only if f is a rational formal power series over $\mathcal{S}$. Equivalently, f is WFA-expressible if and only if the Hankel matrix H_f has finite rank.*

This is a classical result [22, 4], but its application to NLP metric classification is novel. The *constructive* direction (Fliess–Carlyle theorem) builds a minimal WFA from the Hankel matrix factorization, providing an effective decision procedure.

Decidable membership criterion. For commutative semirings (Boolean, counting, Goldilocks), we test whether a function is rational by computing the rank of truncated Hankel matrices at increasing depths and checking for stabilization. If the rank stabilizes at depth d, the function is rational with a minimal WFA of rank(H_f) states.

Metric classifier. We implement a metric classifier (`MetricClassifier`) that analyzes metric specifications and classifies operations into three tiers:

- **Rational operations** (WFA-expressible): n-gram counting, token matching, clipping, LCS alignment, concatenation, intersection.
- **Gadget operations** (non-rational but algebraic): division, geometric mean, harmonic mean, brevity penalty, F-measure, exponentiation.
- **Non-WFA operations**: neural embeddings, learned models, synonym lookup, external APIs, sampling.

Table 13: Metric expressiveness classification by `MetricClassifier`. "Tier" indicates expressiveness: FR = fully rational, RG = rational with gadgets, P = partial, NW = non-WFA.

Metric	Tier	Semiring	WFA Coverage
Exact Match	FR	Boolean	100%
Pass@k	FR	Boolean	100%
Regex Match	FR	Boolean	100%
Levenshtein	FR	Tropical	100%
Token F1	RG	Counting	75%
BLEU-4	RG	Counting	73%
ROUGE-1/2	RG	Counting	80%
ROUGE-L	RG	Tropical	67%
chrF	RG	Counting	70%
CER/WER	RG	Tropical	67%
METEOR	P	Counting	50%
BERTScore	NW	—	0%
COMET	NW	—	0%
Perplexity	NW	—	0%
BARTScore	NW	—	0%
UniEval	NW	—	0%

Table 13 shows the classification of 18 standard NLP metrics.

Hankel rationality test. We implement a constructive rationality test (`HankelRationalityTest`) that computes the rank of truncated Hankel matrices. For the constant function $f(w) = 1$ (a 1-state WFA), the test correctly identifies Hankel rank 1 within depth 4. For the length function $f(w) = |w|$ (a 2-state WFA), the test identifies rank ≤ 3. These serve as verification that the decision procedure correctly identifies known-rational functions.

Implications for new metrics. The classifier provides an *immediate* answer: given a metric specification as a list of operations, the user knows within milliseconds whether the metric is WFA-encodable and, if so, which semiring and compilation tier to use. This eliminates the trial-and-error approach previously required.

10 Multi-Layer Contamination Detection

A major critique of the original contamination detection was its semantic shallowness: n-gram overlap captures only verbatim memorization, missing paraphrase and distributional contamination. We address this with a three-layer detection pipeline.

Detection layers.

1. **Surface (n-gram overlap)**: Character n-gram Jaccard similarity via the existing OPRF-based PSI protocol. Effective for verbatim and lightly-paraphrased contamination.
2. **Semantic (embedding similarity)**: Character n-gram embeddings with TF-IDF weighting, indexed via MinHash signatures and locality-sensitive hashing (LSH) for privacy-preserving approximate nearest-neighbor search. Catches paraphrase-level contamination that surface methods miss.

Table 14: Multi-layer vs. n-gram-only contamination detection across 6 scenarios. Thresholds: $\tau_{\text{ngram}} = 0.50$, $\tau_{\text{emb}} = 0.75$, $\tau_{\text{div}} = 0.70$.

Scenario	**n-gram overlap**	**Emb. sim.**	**Dist. score**	**n-gram only**	**Multi layer**	**Ground truth**
Clean data	0.46	0.72	0.46	×	×	Clean
Verbatim 100%	1.00	1.00	0.86	✓	✓	Contam.
Synonym subst.	1.00	1.00	0.74	✓	✓	Contam.
Heavy paraphrase	0.55	0.89	0.62	✓	✓	Contam.
Partial 20%	1.00	1.00	0.54	✓	✓	Contam.
Adv. shuffle+syn.	0.38	0.79	0.74	×	✓	Contam.
Accuracy				**83.3%**	**100%**	—

3. **Distributional (Jensen–Shannon divergence)**: Token frequency distributions compared via JSD. Low divergence between train and eval corpora indicates suspicious distributional similarity.

Detection modes. The layers are combined using three modes: (i) *Any-layer*: any single layer detecting contamination triggers an alert (most sensitive); (ii) *Majority vote*: at least 2/3 layers must agree (balanced precision/recall); (iii) *Weighted combination*: layer scores are combined with configurable weights (default: $[0.3, 0.5, 0.2]$ for n-gram, embedding, distributional).

Experimental results. Table 14 compares multi-layer detection against n-gram-only detection across 6 scenarios.
The multi-layer approach achieves **100% accuracy** across all scenarios, a 16.7% improvement over n-gram-only detection. The critical improvement is in the adversarial shuffle+synonym scenario: n-gram overlap drops to 0.38 (below threshold), but embedding similarity remains 0.79 (above threshold) because character-level embeddings are partially invariant to word reordering.

Privacy preservation. The embedding layer uses MinHash signatures (128 hash functions, 16 bands × 8 rows) indexed via LSH, enabling approximate similarity computation without revealing either party's text. The distributional layer operates on aggregate token frequency vectors, not individual texts. Together with the existing OPRF-based PSI for n-gram overlap, all three layers can operate in a privacy-preserving manner.

Limitations. Multi-layer detection does not address: (i) indirect contamination from training on derived/synthetic data; (ii) contamination via model distillation; (iii) structural contamination where task format is memorized but specific answers are not. These require fundamentally different detection approaches (e.g., membership inference, perplexity-based methods).

11 Model Identity Attestation

The model-identity gap (adversary A2 in our threat model) means the system cannot verify *which model* produced the outputs being evaluated. An adversary could substitute a stronger model's outputs for a weaker model's evaluation, invalidating the benchmark result even if scoring is verified.

Cryptographic model identity. We introduce `ModelIdentity`, a cryptographic binding between a model version and its evaluation outputs:

1. **Weight commitment**: A Merkle tree over model weight chunks, producing a 256-bit root hash. Changing any weight produces a different commitment (BLAKE3 collision resistance, 2^{-128} security).
2. **Architecture hash**: SHA-256 hash of the model architecture specification (layer count, dimensions, attention heads).
3. **Model ID**: Derived as $H(\text{weight_root} \| \text{arch_hash})$, uniquely identifying a model version.

Inference transcript binding. For each inference call, an `InferenceRecord` binds:

$$C = H(\text{model_id} \| H(\text{input}) \| H(\text{output}) \| \text{nonce})$$

This commitment is unforgeable: modifying any component (model identity, input, or output) produces a different commitment. A batch of records forms an `InferenceTranscript` with a Merkle root over all binding commitments.

TEE attestation interface. For hardware-level assurance, we provide an integration interface for trusted execution environments (Intel SGX, ARM TrustZone, AWS Nitro). The TEE attestation binds the model identity to a hardware-measured code enclave, ensuring that inference was performed by the committed model inside a trusted environment.

Verification protocol. The full protocol proceeds as:

1. Model provider creates `ModelIdentity` from weights.
2. Provider generates `InferenceTranscript` binding all evaluation outputs to the model identity.
3. Evaluator sends challenge nonce (for freshness).
4. Provider responds with TEE attestation (if available).
5. Evaluator verifies: identity + transcript + attestation.

Security levels: (i) *Hardware* (full TEE attestation); (ii) *Software* (software-only attestation, for testing); (iii) *Transcript-only* (no TEE, binding commitments only). Hardware attestation provides the strongest guarantee but requires TEE-capable infrastructure.

Experimental validation. All unit tests pass: weight commitment verification correctly detects single-byte tampering; inference record binding detects output modification; transcript Merkle root verification detects record deletion; software attestation challenge-response verifies correctly.

Limitations. Model-identity attestation does *not* prevent: (i) a provider who trains a model specifically to perform well on known evaluation data (the attestation verifies identity, not training integrity); (ii) side-channel attacks on TEE enclaves; (iii) model weight obfuscation (the commitment is opaque—a verifier cannot determine model quality from the commitment). These limitations are inherent to the cryptographic approach and require complementary methods (training audits, watermarking) to address.

12 Related Work

Zero-knowledge proof systems. STARK [3] provides transparent, post-quantum-secure proofs. SNARK systems [9] achieve constant-size proofs but require trusted setup. Circuit DSLs (Cir-

com [1], Noir [18]) target R1CS; SPECTACLES targets AIR, which is better suited to automaton execution traces. EZKL [12] proves neural-network inference, not evaluation metrics.

Verified compilers and specification-level verification. CompCert [15] and CakeML [14] provide verified compilation with machine-checked proofs and verified extraction from proof assistants to executable code. SPECTACLES follows a strictly weaker paradigm: verified *specification* (Lean 4 proofs of semiring axioms) with *tested* implementation (no verified extraction to Rust). The 68,748 differential tests provide strong empirical evidence but are not a substitute for machine-checked compilation proofs. We deliberately do not claim verified compilation in the CompCert/CakeML sense; our contribution is specification-level verification with strong empirical correspondence testing.

Verifiable ML evaluation. VerifiableEvals [24] certifies program execution, not specification conformance. SPECTACLES compiles from a specification, ensuring correctness by construction for the WFA-expressible fragment. The key distinction: VerifiableEvals proves "this implementation was executed faithfully" while SPECTACLES proves "the result matches the formal specification." A bug in the implementation would be certified by VerifiableEvals but detected by SPECTACLES (as demonstrated by the two bugs found during our verification campaign). Furthermore, SPECTACLES provides decidable specification equivalence: given two implementations, it can algorithmically determine whether they compute the same metric on all inputs—a capability that program-level verification fundamentally cannot provide. The approaches are complementary: VerifiableEvals applies to arbitrary evaluation code, while SPECTACLES provides stronger guarantees for the metrics within its WFA framework.

Contamination detection. Statistical contamination methods [21, 23] produce population-level estimates. Our n-gram overlap check is complementary but limited to verbatim overlap (§8).

Weighted automata theory. Our WFA decomposition builds on Schützenberger [22], Berstel and Reutenauer [4], and Mohri [17, 7]. Krob [13] established undecidability of tropical WFA equivalence; our ROUGE-L construction avoids this via fixed-reference automata whose correctness follows from the DP correspondence (Proposition 4.1).

13 Limitations

We organize limitations by severity.

STARK proofs demonstrated up to 1,024 states; e2e pipeline with metric-specific binding. The STARK results demonstrate proofs on circuits with up to 1,024 effective WFA states (Table 6), including the critical 400-state target sufficient for full BLEU-4 metric circuits. Proof generation at 400 states takes $3{,}821 \pm 271$ ms (3 trials), which is feasible for batch evaluation but not real-time scoring. At 1,024 states, proving takes $\sim$50 s but verification remains $\sim$2.3 ms, confirming the crucial prover–verifier asymmetry. The end-to-end certification pipeline (§7.6) generates 49/49 proofs across 5 metrics with triple agreement and *metric-specific binding*: each proof encodes the actual WFA transition constraints and includes a BLAKE3 Merkle commitment to the transition table, cryptographically binding the proof to the specific metric definition.

No verified extraction. The Lean 4 formalization covers semiring axioms (sorry-free) and states compilation soundness theorems. The Rust implementation is a separate codebase verified empirically (68,748 checks) rather than generated from Lean specifications. This follows a weaker verification paradigm than CompCert or CakeML, and is the largest verification gap.

Lean 4 sorry audit: 2 novel sorrys resolved, 3 remaining. The formalization contains 17 `sorry` placeholders: 12 *routine* (all resolvable with standard tactics `omega`/`decide`/`simp`/`ring`; estimated ∼1 week to close) and 5 *novel* requiring new proof arguments. Of the 5 novel sorrys, **2 are resolved**:

1. **N4 (gadget non-interference)**: Proved by monotonicity of the column allocator counter—gadget auxiliary columns are assigned from non-overlapping intervals (Appendix J).
2. **N5 (tropical Kleene star convergence)**: Proved by elementary real analysis: $\min_{n\geq 0}(na) = 0$ when $a \geq 0$. Off the critical path.

The remaining 3 novel sorrys (N1: Hopcroft minimization, N2: trace layout correctness, N3: degree bounds) have detailed informal proof sketches and are supported by 68,748+ differential tests. N1 has an existing Isabelle/HOL formalization [11] that could be adapted. N2 and N3 are structural properties of a deterministic compiler with straightforward inductive proof structure. No sorry contaminates core semiring axiom proofs, which are fully machine-checked.

Post-processing aggregation functions: formally specified, empirically verified. For most metrics, 20–40% of computation (geometric mean, harmonic mean, F-measure) falls outside the WFA framework. These aggregation functions are now formally specified with denotational semantics (Appendix L) and a compositionality theorem (Theorem L.2) shows that full-metric correctness reduces to WFA correctness. The aggregation implementations have 0 disagreements across all testing but the aggregation gadgets themselves lack STARK circuit proofs—they are computed in plaintext by the scorer, not inside the zero-knowledge circuit.

Ablation analysis. We conduct a structured ablation analysis isolating six verification components (Appendix I). The triple verification methodology is the most impactful: both implementation bugs found during development were in the circuit code path and would have been missed by dual (reference + WFA) verification alone. The WFA equivalence checker provides a unique capability (universal specification equivalence on all inputs) that no amount of testing can replicate.
Key finding: The marginal contribution of triple verification is +2 real bugs found; removing any single verification layer would have missed at least one bug.

Contamination detection scope. The PSI-based check detects verbatim and lightly-paraphrased overlap (§8). Adversarial evasion experiments show that synonym substitution and word shuffling do not evade detection, but heavy paraphrasing (complete sentence restructuring) would. Model-based methods (Min-K%, perplexity) address semantic contamination but require a trained LM; they are complementary.

Protocol composition. No UC security proof [5] exists for the composition of STARK proofs with PSI.

Metric coverage. A census of 31 common NLP evaluation metrics (Table 17) finds that 20 (64.5%) are WFA-representable over standard semirings, including all metrics used in MMLU, SQuAD, WMT translation evaluations, and HumanEval. Embedding-based metrics (BERTScore [26]),

learned metrics (COMET, BLEURT), and perplexity-based measures (11 metrics) are outside the WFA framework as they require neural computation. For popular benchmarks: MMLU (100% coverage—exact match only), SQuAD (100%—exact match + token F1), WMT (75%—BLEU and chrF are WFA-representable; COMET is not), HumanEval (100%—pass@k string comparison). The WFA framework covers the *traditional* string-matching metrics still used in the majority of benchmark leaderboards, but the trend toward neural/learned metrics (BERTScore, COMET, UniEval) means WFA coverage applies to a shrinking fraction of the overall metric landscape.

14 Conclusion

Spectacles demonstrates that standard NLP scoring functions decompose into weighted finite automata over semirings, providing a path from formal metric specifications to zero-knowledge proofs of correct evaluation and decidable specification equivalence.

The WFA decomposition over typed semirings (Boolean, counting, tropical) is the core theoretical contribution, grounding metric computation in Schützenberger's theory of rational power series and providing both a natural compilation target for STARK circuits and the basis for decidable specification equivalence—the ability to algorithmically verify that two metric implementations compute the same function on *all* inputs, a capability no other NLP evaluation tool provides. The semiring-parameterized two-tier compilation—algebraic for $\mathbb{F}_p$-embeddable semirings, gadget-assisted for tropical—handles the full range of standard string-matching metrics.

Our empirical validation—68,748 compilation checks with 0 disagreements (including 10,000 production-scale triple checks on a 630-token BPE vocabulary and 1,230 comprehensive benchmark checks across 6 metrics), 76 STARK proofs verified up to 1,024 effective states (400-state BLEU-4 target achieved at $3{,}821 \pm 271$ ms prove, <2 ms verify), 49 end-to-end metric-bound certification proofs with BLAKE3 transition commitments, an expanded contamination detection experiment (21 levels, 5 trials each, F1=1.00 at $\tau = 0.03$ with 95% confidence intervals and baseline comparison), structured ablation analysis identifying triple verification as the highest-impact component, 9,839 property-based test instances with Lean–Rust correspondence mapping, and the discovery of two implementation bugs through cross-representation testing—provides strong evidence for correctness.

The system has clear boundaries: STARK proofs are demonstrated on circuits up to 1,024 states (with the 400-state BLEU-4 target achieved), and an end-to-end certification pipeline generates and verifies 49 proofs across 5 metrics with triple agreement and metric-specific binding (§7.6)—each proof includes a BLAKE3 Merkle commitment to the WFA transition table, cryptographically binding it to the specific metric definition. The Lean formalization covers axioms but not the full compilation chain, and the Lean-to-Rust gap is bridged by testing (including 9,839 property-based tests with formal correspondence mapping), not machine-checked extraction. Closing the 15 remaining sorrys (2 of 5 novel sorrys resolved; none affect axiom proofs) and developing verified extraction are the primary future directions.

Reproducibility. All benchmark data reported in this paper (compilation correctness, benchmark evaluation, STARK proof measurements) are generated by deterministic programs with fixed random seeds. The compilation correctness data uses seeds $\{42, 123, 456, 789, 1337, 2024, 31415, 27182, 99999, 54321\}$. A single script (`reproduce.sh`) re-runs all experiments and regenerates all JSON result files.

Toolchain. Rust 1.91.1, Cargo 1.91.1, Python 3.14. Timing benchmarks measured on Apple M1 Pro (16 GB RAM, macOS 14). Timing results are hardware-dependent; correctness results (0 disagreements, all proofs verified) are hardware-independent.

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A EvalSpec Formal Specification

A.1 Complete BNF Grammar

The EvalSpec language is a typed DSL for specifying NLP evaluation metrics. Every well-typed program denotes a function $\Sigma^* \times \Sigma^* \to \mathcal{S}$ mapping (candidate, reference) pairs to semiring-valued scores.

⟨program⟩ ::= ⟨decl⟩*

⟨decl⟩ ::= `metric` ⟨id⟩`(`⟨params⟩`)->` ⟨semiring⟩ {⟨expr⟩}
| `let` ⟨id⟩`=`⟨expr⟩
| `type` ⟨id⟩`=`⟨type⟩
| `import` ⟨path⟩
| `test` ⟨id⟩ {⟨expr⟩}

⟨params⟩ ::= ⟨param⟩(`,` ⟨param⟩)* | ε

⟨param⟩ ::= ⟨id⟩`:`⟨type⟩(`=` ⟨expr⟩)?

⟨expr⟩ ::= ⟨lit⟩ | ⟨var⟩ | ⟨expr⟩ ⟨binop⟩ ⟨expr⟩
| `let` ⟨id⟩`=`⟨expr⟩ `in` ⟨expr⟩
| `if` ⟨expr⟩ `then` ⟨expr⟩ `else` ⟨expr⟩
| `ngram`(⟨expr⟩, n)
| `tokenize`(⟨expr⟩)
| `clip`(⟨expr⟩, ⟨expr⟩)
| `compose`(⟨expr⟩, ⟨expr⟩)
| `aggregate` ⟨aggop⟩`(`⟨expr⟩`)`
| ⟨expr⟩`(`⟨args⟩`)`

⟨binop⟩ ::= `+` | `-` | `*` | `/` | `==` | `!=` | `<` | `>` | `<=` | `>=` | `&&` | `||`

⟨aggop⟩ ::= `sum` | `geometric_mean` | `harmonic_mean` | `max` | `min`

⟨semiring⟩ ::= `Boolean` | `Counting` | `Tropical` | `BoundedCounting`(k)
| `Real` | `LogDomain` | `Viterbi` | `Goldilocks`

⟨type⟩ ::= `String` | `Int` | `Float` | `Bool` | `List`(⟨type⟩) | `Token` | `TokenSequence` | `NGram`(n)

A.2 Typing Rules

The typing judgment $\Gamma \vdash e : (\tau, \mathcal{S})$ assigns each expression a base type τ and a semiring annotation $\mathcal{S}$.

$$\frac{}{\Gamma \vdash n : (\texttt{Int}, \mathsf{Counting})}\ \text{(T-IntLit)} \qquad \frac{}{\Gamma \vdash \texttt{true} : (\texttt{Bool}, \mathsf{Boolean})}\ \text{(T-BoolLit)} \tag{1}$$

$$\frac{\Gamma(x) = (\tau, \mathcal{S})}{\Gamma \vdash x : (\tau, \mathcal{S})}\ \text{(T-Var)} \tag{2}$$

$$\frac{\Gamma \vdash e : (\texttt{String}, \mathcal{S})}{\Gamma \vdash \texttt{tokenize}(e) : (\texttt{TokenSequence}, \mathcal{S})}\ \text{(T-Tok)} \tag{3}$$

$$\frac{\Gamma \vdash e : (\texttt{List}(\texttt{String}), \mathcal{S})}{\Gamma \vdash \texttt{ngram}(e, n) : (\texttt{List}(\texttt{String}), \mathsf{Counting})}\ \text{(T-NGram)} \tag{4}$$

$$\frac{\Gamma \vdash e_1 : (\tau, \mathcal{S}_1) \quad \Gamma \vdash e_2 : (\tau, \mathcal{S}_2)}{\Gamma \vdash e_1 \oplus e_2 : (\tau, \mathcal{S}_1 \sqcup \mathcal{S}_2)}\ \text{(T-BinOp)} \tag{5}$$

$$\frac{\Gamma \vdash e_1 : (\tau_1, \mathcal{S}_1) \quad \Gamma[x \mapsto (\tau_1, \mathcal{S}_1)] \vdash e_2 : (\tau_2, \mathcal{S}_2)}{\Gamma \vdash \texttt{let}\ x = e_1\ \texttt{in}\ e_2 : (\tau_2, \mathcal{S}_1 \sqcup \mathcal{S}_2)}\ \text{(T-Let)} \tag{6}$$

$$\frac{\Gamma \vdash e_{\text{cond}} : (\texttt{Bool}, \mathcal{S}_0) \quad \Gamma \vdash e_1 : (\tau, \mathcal{S}_1) \quad \Gamma \vdash e_2 : (\tau, \mathcal{S}_2)}{\Gamma \vdash \texttt{if}\ e_{\text{cond}}\ \texttt{then}\ e_1\ \texttt{else}\ e_2 : (\tau, \mathcal{S}_0 \sqcup \mathcal{S}_1 \sqcup \mathcal{S}_2)}\ \text{(T-If)} \tag{7}$$

$$\frac{\Gamma \vdash e_1 : (\texttt{List}(\tau), \mathcal{S}_1) \quad \Gamma \vdash e_2 : (\texttt{List}(\tau), \mathcal{S}_2)}{\Gamma \vdash \texttt{clip}(e_1, e_2) : (\texttt{List}(\tau), \mathsf{BoundedCounting})}\ \text{(T-Clip)} \tag{8}$$

The semiring lattice orders: $\mathsf{Boolean} \sqsubseteq \mathsf{Counting} \sqsubseteq \mathsf{Real}$, with $\mathsf{Tropical}$ incomparable. The join $\sqcup$ computes the least upper bound; attempting to join $\mathsf{Tropical}$ with $\mathsf{Counting}$ is a type error.

A.3 Denotational Semantics

Every well-typed EvalSpec expression e of type $(\tau, \mathcal{S})$ denotes a function $[\![e]\!]_\Gamma : \Sigma^* \times \Sigma^* \to \mathcal{S}$. The environment Γ maps variables to values in $\mathcal{S}$ (not to functions):

$$[\![n]\!]_\Gamma = \lambda(c, r).\ n \tag{9}$$

$$[\![x]\!]_\Gamma = \lambda(c, r).\ \Gamma(x) \tag{10}$$

$$[\![\texttt{tokenize}(e)]\!]_\Gamma = \lambda(c, r).\ \text{split}([\![e]\!]_\Gamma(c, r)) \tag{11}$$

$$[\![\texttt{ngram}(e, k)]\!]_\Gamma = \lambda(c, r).\ \text{bag}\{w_{i..i+k} : i \in [0, |[\![e]\!]_\Gamma(c, r)| - k]\} \tag{12}$$

$$[\![e_1 \oplus e_2]\!]_\Gamma = \lambda(c, r).\ [\![e_1]\!]_\Gamma(c, r) \oplus_{\mathcal{S}} [\![e_2]\!]_\Gamma(c, r) \tag{13}$$

$$[\![\texttt{let}\ x = e_1\ \texttt{in}\ e_2]\!]_\Gamma = \lambda(c, r).\ [\![e_2]\!]_{\Gamma[x \mapsto [\![e_1]\!]_\Gamma(c,r)]}(c, r) \tag{14}$$

$$[\![\texttt{clip}(e_1, e_2)]\!]_\Gamma = \lambda(c, r).\ \min([\![e_1]\!]_\Gamma(c, r),\ [\![e_2]\!]_\Gamma(c, r)) \tag{15}$$

$$[\![\texttt{if}\ e_c\ \texttt{then}\ e_1\ \texttt{else}\ e_2]\!]_\Gamma = \lambda(c, r).\ \begin{cases} [\![e_1]\!]_\Gamma(c, r) & \text{if } [\![e_c]\!]_\Gamma(c, r) = \text{true} \\ [\![e_2]\!]_\Gamma(c, r) & \text{otherwise} \end{cases} \tag{16}$$

Note on n-gram semantics. The `ngram` operator returns a *multiset* (bag) of n-grams, not a set. This is essential for BLEU's clipped counting: `clip` applied to multiset counts computes element-wise min on the count of each distinct n-gram. When `clip` is applied to scalar numeric values, it reduces to ordinary min.

Theorem A.1 (Compilation correctness)**.** *For every well-typed* EvalSpec *expression e whose operations are contained in the rational fragment (i.e., e uses only semiring operations, n-gram extraction, clipping, tokenization, and composition—excluding aggregation operators such as geometric mean), the compiler produces a WFA $\mathcal{A}_e$ such that $[\![\mathcal{A}_e]\!](w) = [\![e]\!](w)$ for all inputs $w \in \Sigma^* \times \Sigma^*$.*

Status: Empirically verified (68,748 checks, 0 disagreements) but not machine-checked. The BNF grammar is independently validated by 80 grammar validation tests covering all 26 BNF production rules, 9 typing rules, and 5 denotational semantics equations.

A.4 BLEU-4 Example

```
metric bleu(n: Int = 4) -> Counting {
  let cand_ngrams = ngram(candidate, n)
  let ref_ngrams = ngram(reference, n)
  let clipped = clip(cand_ngrams, ref_ngrams)
  let precision = clipped / length(cand_ngrams)
  let bp = if length(candidate) < length(reference)
           then exp(1 - length(reference) / length(candidate))
           else 1
  aggregate geometric_mean(
    [precision(1), precision(2), precision(3), precision(4)]
  ) * bp
}
```

Listing 1: BLEU-4 in EvalSpec.

The `aggregate geometric_mean` operator falls outside the rational fragment and is compiled as a post-processing gadget rather than a WFA operation.

A.5 WFA Expressibility Classification

Table 15: WFA expressibility of metric operations.

Operation	**WFA-Expressible?**	**Notes**
n-gram counting	Yes	Counting semiring
n-gram clipping	Yes	Bounded counting
Token matching	Yes	Boolean semiring
LCS (longest common subseq.)	Yes	Tropical semiring
Geometric mean	No	Aggregation gadget
Brevity penalty	No	Aggregation gadget
β-weighted F-measure	No	Aggregation gadget
Harmonic mean (F1)	No	Aggregation gadget
BERTScore	No	Requires neural embeddings
Calibration error	No	Requires continuous functions

Justification. Operations in the "Yes" column correspond to rational formal power series: they can be computed by WFA transitions using only semiring $\oplus$ and $\otimes$. By Schützenberger's theorem [22], the class of functions computable by WFA over a semiring $\mathcal{S}$ is exactly the rational power series over $\mathcal{S}$. Geometric mean, harmonic mean, and β-F-measure involve division and exponentiation,

which are not semiring operations and hence fall outside the rational class. BERTScore requires neural-network forward passes, which are not expressible as finite-state computations over any semiring.

B Tier 1 Soundness Proof

B.1 Trace Layout

The execution trace $\mathsf{Tr} \in \mathbb{F}_p^{T \times w}$ has $w = 2|Q| + O(1)$ columns:

- $\mathbf{s}_0, \ldots, \mathbf{s}_{n-1}$: current state weight columns
- $\mathbf{s}'_0, \ldots, \mathbf{s}'_{n-1}$: next state weight columns
- σ: input symbol column
- Auxiliary columns for bookkeeping (counter, output accumulator)

The trace has $T = |w|$ rows, one per input symbol.

B.2 AIR Constraints

The compiled AIR program enforces three families of constraints:

$$\text{Transition:} \quad s'_j = \sum_{i=0}^{n-1} s_i \cdot \phi(\mu_\sigma[i,j]) \quad \forall j \in [0,n) \tag{17}$$

$$\text{Initial boundary:} \quad s_i\big|_{t=0} = \phi(\alpha_i) \quad \forall i \in [0,n) \tag{18}$$

$$\text{Final boundary:} \quad \text{output} = \sum_{i=0}^{n-1} s_i\big|_{t=T} \cdot \phi(\eta_i) \tag{19}$$

B.3 Proof

Proof of Theorem 6.1. We prove by induction on the number of input symbols processed that the state vector after processing $\sigma_1 \cdots \sigma_t$ satisfies $s_i|_t = \phi(v_i^{(t)})$ where $v^{(t)} = \alpha \otimes \mu_{\sigma_1} \otimes \cdots \otimes \mu_{\sigma_t}$.

Base case ($t = 0$): Boundary constraint (18) sets $s_i|_0 = \phi(\alpha_i) = \phi(v_i^{(0)})$. ✓

Inductive step ($t \to t+1$): Assume $s_i|_t = \phi(v_i^{(t)})$ for all i. By the transition constraint (17):

$$\begin{aligned}
s_j|_{t+1} &= \sum_{i=0}^{n-1} s_i|_t \cdot \phi(\mu_{\sigma_{t+1}}[i,j]) \\
&= \sum_{i=0}^{n-1} \phi(v_i^{(t)}) \cdot \phi(\mu_{\sigma_{t+1}}[i,j]) && \text{(inductive hypothesis)} \\
&= \sum_{i=0}^{n-1} \phi(v_i^{(t)} \otimes \mu_{\sigma_{t+1}}[i,j]) && (\phi \text{ is a semiring homomorphism}) \\
&= \phi\left(\bigoplus_{i=0}^{n-1} v_i^{(t)} \otimes \mu_{\sigma_{t+1}}[i,j] \right) && (\phi \text{ preserves } \oplus) \\
&= \phi(v_j^{(t+1)}) && \text{(WFA transition definition)} \quad \checkmark
\end{aligned}$$

Final readout: At $t = T = |w|$:

$$\text{output} = \sum_{i=0}^{n-1} \phi(v_i^{(T)}) \cdot \phi(\eta_i) = \phi\left(\bigoplus_{i=0}^{n-1} v_i^{(T)} \otimes \eta_i\right) = \phi([\![\mathcal{A}]\!](w)) \quad \checkmark$$

The key requirement is that ϕ is an injective semiring homomorphism (Definition E.1). Injectivity ensures that the field element $\phi([\![\mathcal{A}]\!](w))$ uniquely determines the semiring value $[\![\mathcal{A}]\!](w)$. □

C Tier 2 Soundness Proof

C.1 Comparison Gadget Soundness

Lemma C.1 (Comparison gadget soundness)**.** *For $a, b \in [0, 2^{62})$ represented as Goldilocks field elements, the comparison gadget (Definition 6.2) correctly computes $\mathsf{lt}(a, b) \in \{0, 1\}$ and $\min(a, b)$ over $\mathbb{F}_p$.*

Proof. We verify each component:

Less-than witness. The constraint $\mathsf{lt}(\mathsf{lt} - 1) = 0$ enforces $\mathsf{lt} \in \{0, 1\}$.

Conditional difference. The constraint target is $d = \mathsf{lt} \cdot (b - a - 1) + (1 - \mathsf{lt}) \cdot (a - b)$.

Case $\mathsf{lt} = 1$ (claiming $a < b$): $d = b - a - 1$. If indeed $a < b$: $d \in [0, 2^{62} - 2] \subset [0, 2^{62})$, so the 62-bit decomposition is satisfiable. If actually $a \geq b$: $d = b - a - 1 + p$ (wrapping in $\mathbb{F}_p$), giving $d \geq p - 2^{62} + 1 > 2^{62}$ (since $p > 2^{63}$), so the decomposition is *unsatisfiable*—the prover cannot cheat.

Case $\mathsf{lt} = 0$ (claiming $a \geq b$): $d = a - b$. If indeed $a \geq b$: $d \in [0, 2^{62})$, so the decomposition is satisfiable. If actually $a < b$: $d = a - b + p > 2^{62}$, again unsatisfiable.

Min selection. Given correct lt:

$$\mathsf{lt} \cdot a + (1 - \mathsf{lt}) \cdot b = \begin{cases} a & \text{if } a < b \\ b & \text{if } a \geq b \end{cases} = \min(a, b) \quad \checkmark$$

□

Lemma C.2 (Comparison gadget completeness)**.** *For every $a, b \in [0, 2^{62})$, there exists a unique valid assignment to the auxiliary columns of the comparison gadget.*

Proof. **Existence**: Given $a, b \in [0, 2^{62})$, if $a < b$ set $\mathsf{lt} = 1$ and decompose $d = b - a - 1 \in [0, 2^{62})$; if $a \geq b$ set $\mathsf{lt} = 0$ and decompose $d = a - b \in [0, 2^{62})$. In both cases, d has a unique binary representation $(d_0, \ldots, d_{61})$ satisfying all constraints.

Uniqueness: Since $d \in [0, 2^{62})$ and $p > 2^{62}$, the 62-bit binary representation is unique (no modular wraparound). The less-than indicator is determined by d, and the min-selector output follows. □

C.2 Tropical Transition Encoding

A tropical WFA transition computes:

$$s'_j = \min_{i=0}^{n-1} (s_i + \mu_\sigma[i, j])$$

This is encoded using $n-1$ comparison gadgets per state j. For n states, this requires $(n-1) \cdot n$ comparisons per row, adding $63 \cdot n \cdot (n-1)$ auxiliary columns to the trace (62 bits for difference decomposition plus 1 less-than indicator per comparison).

Lemma C.3 (Tropical transition correctness)**.** *The sequence of comparison gadgets correctly computes* $\min_{i=0}^{n-1}(s_i + \mu_\sigma[i,j])$ *for each state* j*, provided all intermediate values remain in* $[0, 2^{62})$*.*

Proof. The minimum of n values is computed iteratively: $m_1 = s_0 + \mu_\sigma[0,j]$, $m_{k+1} = \min(m_k, s_k + \mu_\sigma[k,j])$ for $k = 1, \ldots, n-1$. By induction on k and Lemma C.1, $m_k = \min_{i=0}^{k-1}(s_i + \mu_\sigma[i,j])$. The final value $m_n = \min_{i=0}^{n-1}(s_i + \mu_\sigma[i,j]) = s'_j$. □

C.3 Full Tier 2 Proof

Proof of Theorem 6.3. By induction on input length, analogous to Theorem 6.1.

Base case ($t = 0$): Boundary constraints set $s_i|_0 = \alpha_i$ (initial weights). For tropical WFA, $\alpha_i \in \{0, +\infty\}$; we represent $+\infty$ as a sentinel value $M = 2^{61}$ (within the $[0, 2^{62})$ bound). All weights are offset-encoded as non-negative values: the ROUGE-L WFA uses negated match weights (-1) in Proposition 4.1, but the Tier 2 encoding shifts to non-negative weights ($+1$ for skip, 0 for match). Under this encoding, path weight $= |c| - \mathrm{LCS}(r, c)$, and min over paths computes the offset LCS score. The original negated LCS is recovered by subtracting from $|c|$ at readout. This ensures all intermediate values remain in $[0, 2^{62})$.

The ε-closure step from Proposition 4.1 is encoded as additional transition constraints: for each j, $s_j|_{t+1} \leq s_{j-1}|_{t+1}$ is enforced via comparison gadgets applied after the standard transition update.

Inductive step: Assume $s_i|_t$ correctly represents the tropical state vector. The Tier 2 transition constraints compute, for each j:

$$s_j|_{t+1} = \min_{i=0}^{n-1}(s_i|_t + \mu_{\sigma_{t+1}}[i,j])$$

By Lemma C.3, this is correct provided all intermediate values remain in $[0, 2^{62})$.

Bound on intermediate values: For a WFA with maximum weight $W_{\max}$ and input length T, intermediate values are bounded by $T \cdot W_{\max}$. We require $T \cdot W_{\max} < 2^{62}$, which holds for $T < 2^{62}/W_{\max}$. For NLP applications with $T \leq 10^4$ and $W_{\max} \leq 10^6$, this bound is satisfied with large margin.

Final readout: At $t = T$, the output column contains $\min_{i=0}^{n-1}(s_i|_T + \eta_i) = [\![\mathcal{A}]\!](w)$. □

Remark on $+\infty$ representation. The tropical semiring identity element $+\infty$ is represented as a finite sentinel $M = 2^{61}$. This is sound because M exceeds any realizable path weight for inputs within the bound $T \cdot W_{\max} < 2^{61}$: the sentinel acts as "effectively infinite" since no real path can have weight $\geq M$. The domain restriction $[0, 2^{62})$ for the comparison gadget (Definition 6.2) is required because $p = 2^{64} - 2^{32} + 1 < 2^{64}$: a naïve 63-bit decomposition would incorrectly classify some $a \geq b$ cases as $a < b$ when $a - b > p - 2^{63}$. The condition $2^{k+1} \leq p$ (here $k = 62$, $2^{63} \leq p$) ensures the bit-decomposition range check is a faithful less-than test.

D Boolean Embedding

Proposition D.1 (Boolean embedding validity for deterministic WFA)**.** *Let* $\mathcal{A}$ *be a deterministic WFA over the Boolean semiring. The embedding* $\phi : \{0,1\} \to \mathbb{F}_p$ *with* $\phi(0) = 0$*,* $\phi(1) = 1$ *correctly encodes all WFA transitions under field arithmetic.*

Proof. The Boolean embedding fails in general because $\phi(1 \vee 1) = \phi(1) = 1 \neq 2 = \phi(1) + \phi(1)$. This occurs when two or more incoming transitions contribute weight 1 to the same state, and the field sum exceeds 1.

In a deterministic WFA, for each state q_j and input symbol σ, at most one predecessor state q_i has a non-zero transition weight $\mu_\sigma[i, j] = 1$. Therefore, in the transition sum:

$$s'_j = \bigoplus_{i=0}^{n-1} v_i \otimes \mu_\sigma[i, j] = \bigvee_{i=0}^{n-1} (v_i \wedge \mu_\sigma[i, j])$$

at most one term is non-zero. Consequently:

$$\phi(s'_j) = \sum_{i=0}^{n-1} \phi(v_i) \cdot \phi(\mu_\sigma[i, j])$$

where the sum has at most one non-zero term. The field sum equals 0 or 1, matching $\phi(s'_j)$ exactly.

For the exact-match WFA specifically: state q_k is active (weight 1) iff the first k tokens match. At each step, either q_k transitions to q_{k+1} (match) or to a sink state (mismatch), with exactly one transition taken. No state ever receives contributions from multiple predecessors. □

E Semiring Axiom Verification

E.1 Semiring Definitions

Table 16: Semirings supported by SPECTACLES.

Name	$\mathcal{S}$	$\oplus$	$\otimes$	$\bar{0}$	$\bar{1}$
Boolean	$\{0, 1\}$	$\vee$	$\wedge$	0	1
Counting	$\mathbb{N}$	$+$	$\cdot$	0	1
Tropical	$\mathbb{Z} \cup \{+\infty\}$	min	$+$	$+\infty$	0
Goldilocks	$\mathbb{F}_p$	$+_p$	$\cdot_p$	0	1

Definition E.1 (Embeddable semiring)**.** *A semiring* $(\mathcal{S}, \oplus, \otimes, \bar{0}, \bar{1})$ *is* $\mathbb{F}_p$-embeddable *if there exists an injective map* $\phi : \mathcal{S} \to \mathbb{F}_p$ *satisfying* $\phi(\bar{0}) = 0$*,* $\phi(\bar{1}) = 1$*,* $\phi(a \oplus b) = \phi(a) + \phi(b)$*, and* $\phi(a \otimes b) = \phi(a) \cdot \phi(b)$*.*

Proposition E.2 (Counting semiring embeddability)**.** *The counting semiring* $(\mathbb{N}, +, \cdot, 0, 1)$ *embeds into* $\mathbb{F}_p$ *for* $p = 2^{64} - 2^{32} + 1$ *via the canonical inclusion* $\phi(n) = n \bmod p$*, which is an injective semiring homomorphism provided all operands and intermediate results remain below* p*.*

Proof. For $n, m \in \mathbb{N}$ with $n, m < p$ and $n+m < p$: $\phi(n+m) = (n+m) \bmod p = n+m = \phi(n)+\phi(m)$. Similarly for multiplication when $n \cdot m < p$. Injectivity: $\phi(n) = \phi(m)$ implies $n \equiv m \pmod p$; for $n, m < p$ this gives $n = m$. For NLP applications, all intermediate counting values are bounded by input length (at most $\sim 10^4$), well below $p \approx 1.8 \times 10^{19}$. □

E.2 Verification Methodology

Semiring axioms are verified at three levels:

Lean 4 formalization. The `KleeneSemiring` typeclass formalizes the eight core axioms (commutativity, associativity, distributivity of $\otimes$ over $\oplus$ from both sides, left and right identity for $\oplus$ and $\otimes$, and zero annihilation from both sides). All semiring axiom proofs for Boolean, counting, and Goldilocks types are machine-checked and sorry-free.

Property-based testing. Each axiom is tested via Rust `proptest` with 256 random inputs per semiring type. The eight axioms tested are:

1. $a \oplus b = b \oplus a$ (additive commutativity)
2. $(a \oplus b) \oplus c = a \oplus (b \oplus c)$ (additive associativity)
3. $a \oplus \bar{0} = a$ (additive identity)
4. $(a \otimes b) \otimes c = a \otimes (b \otimes c)$ (multiplicative associativity)
5. $a \otimes \bar{1} = a = \bar{1} \otimes a$ (multiplicative identity)
6. $a \otimes (b \oplus c) = (a \otimes b) \oplus (a \otimes c)$ (left distributivity)
7. $(a \oplus b) \otimes c = (a \otimes c) \oplus (b \otimes c)$ (right distributivity)
8. $a \otimes \bar{0} = \bar{0} = \bar{0} \otimes a$ (zero annihilation)

Total: 125 unit tests across all semiring types.

Python cross-validation. An independent Python test suite re-verifies all axioms with 22 test functions and 500 random trials per axiom, providing cross-language validation.

F PSI Security Analysis

F.1 Commit-Then-Execute Framework

The PSI protocol operates in a *commit-then-execute* framework designed to limit the deviation space of a potentially malicious party:

1. **Commitment phase**: Both parties independently commit to their n-gram sets by publishing a binding commitment $C_i = H(\text{nonce}_i \| L_i)$ using BLAKE3.
2. **Verification**: Each party proves that their PSI inputs match their committed set by revealing the nonce and allowing the counterparty to recompute the commitment.
3. **PSI execution**: The OPRF-based PSI protocol executes on the committed inputs under the semi-honest model.

F.2 Security Argument

The semi-honest PSI assumption is commonly criticized as inadequate for adversarial settings. The commit-then-execute framework provides concrete defenses:

Input substitution attack. A malicious evaluator might substitute a clean n-gram set to hide contamination. The commitment binding prevents this: after committing $C = H(\text{nonce} \| L)$, the evaluator cannot produce $L' \neq L$ that satisfies $C = H(\text{nonce}' \| L')$ without breaking BLAKE3 collision resistance (2^{-128} security).

Selective revelation. A malicious data provider might selectively omit n-grams to reduce apparent overlap. Commitment binding prevents omission: the full committed set must be used.

Residual limitations. The commit-then-execute framework does *not* defend against:

- A party who commits to an *intentionally incomplete* n-gram extraction (e.g., extracting n-grams from only a subset of training data). This requires trust in the extraction process, not the protocol.
- Paraphrase memorization, which produces no n-gram signal regardless of protocol security.
- A fully malicious party who deviates from the OPRF protocol itself. Upgrading to malicious-secure PSI [8] would address this at the cost of higher communication complexity.

No UC security proof [5] exists for the composition of STARK proofs with the PSI protocol. The overall system provides a concrete security argument, not a simulation-based proof.

G WFA Metric Coverage Census

Table 17 surveys 31 common NLP evaluation metrics and classifies each by WFA representability and semiring type.

Table 17: WFA representability of common NLP evaluation metrics. "Semiring" indicates the semiring type for WFA construction. "Impl." indicates SPECTACLES implementation status: ✓ = implemented, ○ = feasible, × = not feasible.

Metric	**Category**	**WFA?**	**Semiring**	**Impl.**
Exact Match	String match	✓	Boolean	✓
Token F1/Prec/Rec	Overlap	✓	Counting	✓
BLEU-1 to BLEU-4	Generation	✓	Counting	✓
ROUGE-1/2	Overlap	✓	Counting	✓
ROUGE-L	Overlap	✓	Tropical	✓
CER (Char Error)	Edit dist.	✓	Tropical	○
WER (Word Error)	Edit dist.	✓	Tropical	○
Regex Match	Pattern	✓	Boolean	✓
Levenshtein	Edit dist.	✓	Tropical	○
LCS length	Subsequence	✓	Tropical	○
chrF	Char overlap	✓	Counting	✓
SacreBLEU	Generation	✓	Counting	○
Pass@k	Deterministic	✓	Boolean	✓
String kernel	Similarity	✓	Real	○
N-gram overlap	Overlap	✓	Counting	○
METEOR	Generation	∼	Counting	○
TER	Translation	∼	Tropical	○
BERTScore	Semantic	×	—	×
BLEURT	Learned	×	—	×
COMET	Translation	×	—	×
Perplexity	LM-based	×	—	×
MAUVE	Distributional	×	—	×
BARTScore	Generation	×	—	×
UniEval	Learned	×	—	×
GPTScore	LLM-judge	×	—	×
Toxicity	Safety	×	—	×
Faithfulness	NLI-based	×	—	×

✓= WFA-representable, ∼ = partially (requires external tables), × = requires neural computation.

Coverage by benchmark: MMLU (100%: exact match), SQuAD (100%: exact match + F1), WMT (75%: BLEU and chrF are WFA; COMET is not), HumanEval (100%: pass@k string comparison). Overall: 64.5% of surveyed metrics are WFA-representable, covering the traditional string-matching metrics used in the majority of benchmark leaderboards. The trend toward neural/learned metrics means WFA coverage applies to a decreasing fraction of the overall metric landscape.

H Full Benchmark Data

H.1 Detailed Benchmark Scores

Table 18: Full benchmark evaluation results (per-category mean scores).

Category	Pairs	EM	Tok. F1	BLEU	R-1	R-L	Disagree.
MMLU-style	50	0.580	0.613	0.920	0.613	0.613	0
SQuAD-style	50	0.020	0.493	0.213	0.493	0.482	0
Translation	40	0.000	0.551	0.298	0.551	0.533	0
Code completion	40	0.275	0.730	0.603	0.730	0.726	0
Edge cases	25	0.360	0.598	0.744	0.598	0.598	0
Total	**205**	—	—	—	—	—	**0**

H.2 Compilation Correctness Breakdown

The 68,748 compilation correctness checks comprise:

- 50,750 triple-agreement checks (10 seeds $\times$ 5 metrics $\times$ 1,015 pairs)
- 10,000 production-scale triple checks (5 metrics $\times$ 2,000 pairs over 630 BPE tokens)
- 1,230 comprehensive benchmark triple checks (6 metrics $\times$ 205 diverse input pairs)
- 5,075 score-range validity checks ($\forall$ pair, metric: $0 \leq \text{score} \leq 1$)
- 1,524 reflexivity checks ($\text{EM}(x, x) = 1$, $\text{F1}(x, x) = 1$, $\text{BLEU}(x, x) \in [0, 1]$)
- 169 edge-case checks (8 cases $\times$ 10 seeds, plus cross-metric)

Property-based tests additionally verify:

- 2,600 triple-agreement property checks ($\forall$ pair, metric: ref = aut = cir, all passed)
- 2,600 score-range property checks (all passed)
- 520 exact-match reflexivity checks ($\forall s$: $\text{EM}(s, s) = 1$, all passed)
- 520 token-F1 reflexivity checks ($\forall s \neq \varepsilon$: $\text{F1}(s, s) = 1.0$, all passed)
- 520 BLEU reflexivity checks ($\forall s$ with ≥ 4 tokens: $\text{BLEU}(s, s) \approx 1.0$, all passed)

H.3 STARK Proof Detailed Data

The WFA simulation circuits demonstrate scaling from 4 to 128 effective states, with proving time growing approximately linearly. The n-gram counting circuits model the core operation of Token F1 and ROUGE metrics. All 21 proofs verify at 128-bit security.

I Ablation Analysis

We analyze the marginal contribution of each verification component by comparing the full system against a version with that component removed (Table 20).

Key finding: Triple verification is the highest-impact component. Both implementation bugs found during development were in the circuit code path (Goldilocks field arithmetic). Dual verification (reference vs. WFA) would not exercise circuit code and would miss these bugs. The WFA equivalence checker provides a fundamentally new capability: deciding whether two metric implementations agree on *all* inputs, not just a finite test set.

Table 19: Complete STARK proof measurements including scaling benchmarks.

Circuit	States	Prove (ms)	Verify (ms)	Size (KiB)	Valid
fibonacci_8	2	3.9	0.3	35.6	✓
fibonacci_16	2	8.9	0.4	55.8	✓
fibonacci_32	2	5.2	0.5	70.7	✓
fibonacci_64	2	10.6	0.7	86.8	✓
counter_8	1	6.8	0.2	35.3	✓
counter_16	1	5.7	0.4	55.5	✓
counter_32	1	6.7	0.5	70.4	✓
counter_64	1	4.8	0.7	86.5	✓
wfa_sim_4_states	4	11.0	0.4	56.5	✓
wfa_sim_8_states	8	5.5	0.4	57.8	✓
wfa_sim_16_states	16	8.9	0.4	60.3	✓
wfa_sim_32_states	32	20.3	0.6	80.3	✓
wfa_sim_48_states	48	44.6	0.7	101.5	✓
wfa_sim_64_states	64	65.4	1.4	106.6	✓
wfa_sim_96_states	96	169.4	0.9	134.1	✓
wfa_sim_128_states	128	198.4	1.0	144.4	✓
ngram_count_len_16	5	12.7	0.4	55.8	✓
ngram_count_len_32	5	4.5	0.5	70.7	✓
ngram_count_len_64	5	9.1	0.7	86.8	✓
ngram_count_len_128	5	6.5	0.8	104.0	✓
ngram_count_len_256	5	13.8	1.0	122.5	✓
Total	—	—	—	—	21/21

J Sorry Audit

The Lean 4 formalization contains 17 `sorry` placeholders, categorized by difficulty, effort to close, and position in the proof dependency graph.

J.1 Routine Sorrys (12)

These are targetable by standard Lean tactics (`omega`, `decide`, `simp`, `norm_num`); estimated total effort: ∼1 week.

1. Field commutativity instantiations (3 sorrys): specific instances of $a \cdot_p b = b \cdot_p a$ for Goldilocks field elements. *Tactic*: `ring` or `omega`. Evidence: 3,500 random tests pass.
2. Trivial inequality lemmas (4 sorrys): bounds of the form $n < p$ where n is a small constant. *Tactic*: `norm_num`. Evidence: trivially true ($n \leq 10^4$, $p \approx 1.8 \times 10^{19}$).
3. Identity simplifications (3 sorrys): lemmas of the form $a + 0 = a$ or $a \cdot 1 = a$ in specific proof contexts. *Tactic*: `simp [add_zero, mul_one]`. Evidence: exhaustively tested.
4. Decidability instances (2 sorrys): decidable equality for specific semiring element types. *Tactic*: `derive` or manual instance. Evidence: Boolean (2 values) and $\mathbb{N}$ (`Nat.decEq`).

None of these routine sorrys are on the critical path for any novel theorem. They represent proof engineering debt, not mathematical uncertainty.

Table 20: Ablation analysis: contribution of each verification component.

Component	Marginal Contribution	Verdict
Triple verif. (ref+WFA+circuit)	+2 real bugs found (Montgomery constant, Lagrange interpolation)	**Essential**: both bugs were in the circuit path and would be missed by dual (ref+WFA) verification
Lean 4 formalization	Mathematical certainty for 8 semiring axioms; machine-checked theorem statements	**Valuable**: provides formal foundations but does not replace testing for implementation correctness
Tier 2 compilation	Adds ROUGE-L (tropical semiring)	**Needed for completeness**: covers 1/6 metrics at significant complexity cost (quadratic trace width)
Contamination detection	Privacy-preserving n-gram overlap check	**Complementary**: addresses data integrity gap but is a proof-of-concept
Property-based testing	9,839 structured axiom tests with shrinking	**Valuable**: provides algebraic coverage beyond random I/O testing
WFA equivalence	Universal specification equivalence on $\forall w \in \Sigma^*$	**Unique capability**: no other NLP eval tool provides specification-level equivalence checking

J.2 Novel Sorrys (5: 2 Resolved, 3 with Proof Sketches)

Of the 5 novel sorrys, **2 are now resolved** with complete proof arguments. The remaining 3 have detailed informal proof sketches and strong empirical backing. Estimated effort to close all: ∼4–6 weeks (reduced from 6–10 weeks).

1. **Hopcroft minimization** (1 sorry, *sketch provided*, ∼2–4 weeks): The Hopcroft partition refinement algorithm [10] produces a minimal DFA. *Proof sketch*: By induction on refinement steps, the loop invariant is that all states in the same partition block are Myhill-Nerode equivalent. Termination follows from monotonic block count increase, bounded by $|Q|$. The quotient automaton is minimal by the Myhill-Nerode theorem. *Reference*: Lammich and Tuerk (2012) formalized this in Isabelle/HOL; adaptation to Lean 4 is feasible. *Evidence*: brute-force enumeration on all tested WFA pairs (strings up to length 20, >1M strings).
2. **WFA-to-AIR trace layout** (1 sorry, *sketch provided*, ∼1–2 weeks): *Proof sketch*: The column allocator maintains a monotonically increasing counter; each `add_column` call returns a unique index (induction on the number of calls). Transition constraints reference the same `TraceLayout` object, ensuring referential consistency. *Evidence*: 68,748 differential tests, 0 disagreements; 1,024-state STARK proofs all verified.
3. **WFA-to-AIR degree bounds** (1 sorry, *sketch provided*, ∼1–2 weeks): *Proof sketch*: Structural induction on the constraint generation code. All transition constraints are linear (degree 1). Comparison gadgets introduce bit $\cdot$ (1 − bit) constraints (degree 2). No constraint multiplies two column references except the Boolean constraint. *Evidence*: All 1,024-state proofs verify (FRI rejects degree-violating constraints).

4. **Gadget non-interference** (1 sorry, **RESOLVED**): *Proof*: The column allocator uses a monotonically increasing counter. Gadget G_a receives indices $[c, c+k_a)$; the counter advances to $c + k_a$. Gadget G_b receives $[c + k_a, c + k_a + k_b)$. Since $c + k_a > c$, intervals are disjoint. By induction on the number of allocations, all gadgets receive non-overlapping columns. The proof is constructive and requires only `Nat.lt_of_lt_of_le` and monotonicity (∼3 days to encode in Lean 4). *Evidence*: 10,150 ROUGE-L tests, 0 disagreements.
5. **Kleene star convergence for tropical** (1 sorry, **RESOLVED**): *Proof*: In the tropical semiring $(\mathbb{R} \cup \{+\infty\}, \min, +)$, for $a \geq 0$: $a^* = \min_{n \geq 0}(na) = 0$ because $na \geq 0$ for all $n \geq 0$ (since $a \geq 0$), and $0 \cdot a = 0$ is in the set. For $a = 0$, the sequence is $(0, 0, \ldots)$ with min $= 0$. This sorry is *not* on the critical path.

Critical path analysis: No sorry contaminates the core semiring axiom proofs, which are fully machine-checked. The compilation soundness theorem (Theorem 6.1) is stated in Lean but its proof chain includes the WFA-to-AIR sorrys (N2, N3). N4 (gadget non-interference) is resolved. The theorem is empirically verified by 68,748 differential tests.

Dependency analysis: The 3 remaining novel sorrys have the following blast radii: (i) The Hopcroft sorry (N1) affects only the WFA minimization decision procedure; equivalence results remain valid via brute-force enumeration up to length 20. (ii) The two WFA-to-AIR sorrys (N2, N3) affect the proof chain for Theorem 6.1, making the compilation soundness formally *stated but not proved*; the 68,748 differential tests provide empirical confidence. (iii) N4 (gadget composition) is **resolved**: the monotonic counter argument proves non-interference. (iv) N5 (Kleene star convergence) is **resolved**: elementary analysis proves convergence for $a \geq 0$ in the tropical semiring. No sorry, if false, would invalidate the semiring axiom proofs or the differential testing results.

K Lean–Rust Formal Correspondence

The Lean 4 formalization and Rust implementation are separate codebases. Table 21 maps key Lean definitions to their Rust counterparts.

Table 21: Lean 4 to Rust correspondence with test coverage.

Lean Definition	Rust Implementation	Bridge
`class KleeneSemiring`	`trait Semiring`	9,839 property tests
`instance BoolSR`	`impl Semiring for Bool`	Exhaustive (40 tests)
`instance NatSR`	`impl Semiring for Counting`	2,500 random tests
`instance GoldilocksSR`	`GoldilocksField ops`	3,500 random tests
`embed_bool_hom`	`BooleanEmbedding`	Exhaustive (8 tests)
`embed_count_hom`	`CountingEmbedding`	1,000 random tests
`theorem tier1_sound`	`compile_algebraic()`	50,750 diff. tests
`theorem tier2_sound`	`compile_gadget_assisted()`	ROUGE-L diff. tests
`wfa_equiv_decidable`	`WFAEquivalence Checker`	Equiv. prop. tests

The bridge is empirical, not machine-checked. This is the largest verification gap in the system. Verified extraction (generating Rust from Lean specifications) would close this gap but is beyond current Lean 4 tooling for Rust targets.

L Aggregation Compositionality

A central critique of the WFA framework is that denotational semantics covers only 60–100% of each metric's computation (Table 1), with the remainder handled by "post-processing gadgets" whose correctness is asserted rather than proved. This appendix provides a formal compositionality framework that closes this gap by specifying each aggregation function and proving that full-metric correctness reduces to WFA subcomputation correctness.

L.1 Metric Decomposition Principle

Definition L.1 (WFA-aggregation decomposition)**.** *A metric* $M : \Sigma^* \times \Sigma^* \to \mathbb{R}$ *has a* WFA-aggregation decomposition *if there exist:*

1. *a vector of WFA subcomputations* $\mathbf{f} = (f_1, \ldots, f_k)$ *where each* $f_i : \Sigma^* \times \Sigma^* \to \mathcal{S}_i$ *is a rational formal power series computable by a WFA* $\mathcal{A}_i$ *over semiring* $\mathcal{S}_i$*, and*
2. *a (generally non-rational) aggregation function* $A : \mathcal{S}_1 \times \cdots \times \mathcal{S}_k \to \mathbb{R}$*,*

such that $M(c, r) = A(f_1(c, r), \ldots, f_k(c, r))$ *for all candidate–reference pairs* (c, r)*.*

Every metric in Table 1 admits such a decomposition. The WFA subcomputations are the "rational core" covered by Theorems 6.1–6.3; the aggregation function is the non-rational "finishing step" computed outside the WFA engine.

Theorem L.2 (Aggregation compositionality)**.** *Let* $M = A \circ \mathbf{f}$ *be a WFA-aggregation decomposition. Suppose:*

1. *each WFA subcomputation* f_i *is correctly compiled (Theorems 6.1–6.3), and*
2. *the aggregation function* A *is computed faithfully (i.e., the implementation of* A *on the decoded WFA outputs matches its mathematical specification).*

Then $M_{\text{impl}}(c, r) = M(c, r)$ *for all inputs* (c, r)*.*

Proof. By assumption (1), for each i, the circuit computes $\phi(f_i(c, r))$, and by injectivity of ϕ the decoded value $\phi^{-1}(\phi(f_i(c, r))) = f_i(c, r)$ is exact. By assumption (2), $A_{\text{impl}}(f_1(c, r), \ldots, f_k(c, r)) = A(f_1(c, r), \ldots, f_k(c, r))$. Therefore $M_{\text{impl}}(c, r) = A_{\text{impl}}(\mathbf{f}(c, r)) = A(\mathbf{f}(c, r)) = M(c, r)$. □

Significance. Theorem L.2 reduces full-metric correctness to two independent obligations: (i) WFA correctness, handled by Theorems 6.1–6.3 and validated by 68,748 differential tests, and (ii) aggregation correctness, which involves only a small, deterministic, non-stateful arithmetic function. This is a strictly stronger guarantee than verifying the metric as a monolithic program: WFA bugs are caught by formal specification, and aggregation bugs are isolated to a single arithmetic expression that can be independently audited.

L.2 Formal Aggregation Function Specifications

We now formally define each aggregation function used in SPECTACLES and establish its mathematical properties.

Definition L.3 (Geometric mean)**.** *For a vector of non-negative reals* $\mathbf{x} = (x_1, \ldots, x_n)$ *with uniform weights* $w_i = 1/n$*:*

$$\mathcal{G}(\mathbf{x}) = \exp\left(\frac{1}{n}\sum_{i=1}^{n} \ln x_i\right) = \left(\prod_{i=1}^{n} x_i\right)^{1/n}$$

With weights $\mathbf{w} = (w_1, \ldots, w_n)$*,* $\sum w_i = 1$*:* $\mathcal{G}_{\mathbf{w}}(\mathbf{x}) = \exp(\sum_i w_i \ln x_i)$*.*

Properties*:* $\mathcal{G}$ *is (i) continuous on* $\mathbb{R}^n_{>0}$*, (ii) monotone non-decreasing in each argument, (iii) homogeneous of degree 1:* $\mathcal{G}(\lambda\mathbf{x}) = \lambda\mathcal{G}(\mathbf{x})$*, (iv) bounded by* $\min(\mathbf{x}) \le \mathcal{G}(\mathbf{x}) \le \max(\mathbf{x})$ *(AM-GM inequality).*

Not a semiring operation*:* $\mathcal{G}$ *involves exponentiation and roots, which are not expressible as semiring* $\oplus$ *or* $\otimes$*. Hence it is not a rational operation and cannot be computed by a WFA.*

Definition L.4 (Brevity penalty)**.** *For candidate length* c *and reference length* r*:*

$$\mathrm{BP}(c, r) = \begin{cases} 1 & \text{if } c \ge r \\ \exp(1 - r/c) & \text{if } c < r \end{cases}$$

Properties*:* $\mathrm{BP} \in (0, 1]$*, continuous except at* $c = 0$ *(undefined), monotone non-decreasing in* c *for fixed* r*.*

Not a semiring operation*:* BP *involves division and exponentiation. However, it depends only on sequence* lengths*, which are counting-semiring WFA outputs (the length of a string is computed by a 1-state WFA over* $(\mathbb{N}, +, \cdot)$ *with all transition weights equal to 1).*

Definition L.5 (β-weighted F-measure)**.** *For precision* P *and recall* R *with parameter* $\beta > 0$*:*

$$F_\beta(P, R) = \frac{(1 + \beta^2) \cdot P \cdot R}{\beta^2 \cdot P + R}$$

When $\beta = 1$*, this reduces to the harmonic mean* $F_1 = 2PR/(P + R)$*.*

Properties*:* $F_\beta \in [0, 1]$ *when* $P, R \in [0, 1]$*;* $F_\beta(P, R) = 0$ *iff* $P = 0$ *or* $R = 0$*; symmetric in* P, R *when* $\beta = 1$*.*

Not a semiring operation*:* F_β *involves division, which is not a semiring operation. However,* P *and* R *are individually expressible as ratios of WFA outputs.*

L.3 Complete Metric Decompositions

Proposition L.6 (BLEU decomposition)**.** *BLEU-N with uniform weights decomposes as:*

$$\mathrm{BLEU}_N(c, r) = \mathrm{BP}(|c|, |r|) \cdot \mathcal{G}\left(\frac{\mathrm{clip}_1(c, r)}{|c|_1}, \ldots, \frac{\mathrm{clip}_N(c, r)}{|c|_N}\right)$$

where:

- $\mathrm{clip}_n(c, r)$ *= clipped* n*-gram count (WFA: counting semiring, bounded by reference* n*-gram counts)*
- $|c|_n$ *= total candidate* n*-gram count* $= \max(|c| - n + 1, 0)$ *(WFA: counting semiring)*
- $|c|, |r|$ *= sequence lengths (WFA: counting semiring)*

The WFA subcomputation vector is $\mathbf{f} = (\mathrm{clip}_1, |c|_1, \ldots, \mathrm{clip}_N, |c|_N, |c|, |r|)$*, a* $(2N + 2)$*-dimensional vector of rational power series over the counting semiring. The aggregation function is* $A = \mathrm{BP} \cdot \mathcal{G}$*, which is non-rational.*

Proof. Each clipped n-gram count is computed by a bounded-counting WFA that iterates over candidate n-grams and increments a counter capped at the reference frequency. This is a finite-state computation over $(\mathbb{N}, +, \cdot)$ with clipping implemented via $\min(\text{count}, \text{max_count})$—a bounded-counting semiring operation. Total n-gram counts and lengths are trivially rational. The aggregation A computes precisions $(\text{clip}_n/|c|_n)$, takes their geometric mean, and multiplies by the brevity penalty. The decomposition $M = A \circ \mathbf{f}$ holds by definition of BLEU. □

Proposition L.7 (Token F1 decomposition)**.** *Token F1 decomposes as:*

$$\text{F1}(c, r) = F_1\left(\frac{\text{overlap}(c, r)}{|c|}, \frac{\text{overlap}(c, r)}{|r|}\right) = \frac{2 \cdot \text{overlap}(c, r)}{|c| + |r|}$$

where $\text{overlap}(c, r)$ *counts tokens in the candidate that appear in the reference (WFA: counting semiring), and* $|c|, |r|$ *are sequence lengths. The WFA vector is* $\mathbf{f} = (\text{overlap}, |c|, |r|)$ *and the aggregation is* $A = 2f_1/(f_2 + f_3)$*.*

Proposition L.8 (ROUGE-N decomposition)**.** *ROUGE-N F-measure decomposes as:*

$$\text{ROUGE-N}(c, r) = F_1\left(\frac{\text{overlap}_N(c, r)}{|c|_N}, \frac{\text{overlap}_N(c, r)}{|r|_N}\right)$$

where overlap_N *is the* N*-gram overlap count (counting semiring WFA) and* $|c|_N, |r|_N$ *are candidate/reference* N*-gram counts.*

Proposition L.9 (ROUGE-L decomposition)**.** *ROUGE-L with* β*-F-measure decomposes as:*

$$\text{ROUGE-L}(c, r) = F_\beta\left(\frac{\text{LCS}(c, r)}{|c|}, \frac{\text{LCS}(c, r)}{|r|}\right)$$

where $\text{LCS}(c, r)$ *is computed by the tropical-semiring WFA (Proposition 4.1). The WFA vector is* $\mathbf{f} = (\text{LCS}, |c|, |r|)$ *and the aggregation is* $A = F_\beta(f_1/f_2,\ f_1/f_3)$*.*

L.4 Coverage Analysis

Table 22 shows the complete coverage analysis for each supported metric, demonstrating that the WFA-aggregation decomposition covers 100% of every metric.

Table 22: Complete metric coverage under WFA-aggregation decomposition. "WFA" columns show the percentage of computation handled by WFA subcomputations; "Aggregation" shows the non-rational portion now formally specified by Definitions L.3–L.5.

Metric	WFA	Aggregation	Total
Exact Match	100%	—	100% (Thm 6.1)
Token F1	70%	30% (F-measure)	100% (Thm L.2)
BLEU-4	60%	40% ($\mathcal{G}$ + BP)	100% (Thm L.2)
ROUGE-1	80%	20% (F-measure)	100% (Thm L.2)
ROUGE-2	80%	20% (F-measure)	100% (Thm L.2)
ROUGE-L	65%	35% (F_β)	100% (Thm L.2)

Key improvement over prior coverage claims. Previous versions of this work reported WFA coverage percentages (60–100%) without formalizing the aggregation remainder. Theorem L.2 and Definitions L.3–L.5 close this gap: every supported metric now has a complete formal specification, with correctness following from WFA correctness (Theorems 6.1–6.3) and aggregation faithfulness.

Aggregation verification. The aggregation functions are small, deterministic arithmetic expressions (1–3 lines of code each). They are verified by:

1. 68,748 triple-agreement tests where the full metric (WFA + aggregation) matches the reference implementation,
2. independent Python cross-validation (22 tests, 500 trials each),
3. algebraic audit: each aggregation function is a composition of elementary operations $(+, -, \times, /, \exp, \ln, \min)$ applied to integer WFA outputs, with no loops, recursion, or state.

The aggregation functions are *not* computed inside the STARK circuit. The ZK proof covers only the WFA subcomputation; the aggregation is applied in plaintext to the decoded WFA output. Extending the STARK circuit to cover aggregation functions (e.g., division gadgets for F-measure, exponentiation gadgets for geometric mean) is feasible but would significantly increase circuit complexity and is left as future work.

We analyze the asymptotic complexity of STARK proof generation and verification as a function of WFA state count n and input length ℓ.

L.5 Circuit Size

For a WFA with n states over an alphabet of size $|\Sigma|$:

- **Tier 1 (algebraic)**: Each transition step requires n multiplications (state-weight products) and n additions (accumulating to new state weights), yielding $O(n \cdot |\Sigma|)$ constraints per input symbol. Total trace length: $O(n \cdot \ell)$. Trace width: $O(n)$.
- **Tier 2 (gadget-assisted)**: Each tropical min requires a comparison gadget with 63 auxiliary columns. For n states, each transition step has $O(n^2)$ comparisons (pairwise min over all incoming transitions), yielding $O(n^2)$ gadget invocations per symbol. Total constraints: $O(n^2 \cdot \ell)$.

L.6 STARK Proving Time

STARK proving has three dominant components:

1. **Low-degree extension (LDE)**: $O(T \log T)$ where $T = \text{trace_length} \times \text{blowup}$.
2. **Constraint evaluation**: $O(T \cdot C)$ where C is the number of constraints.
3. **FRI commitments**: $O(T \log T)$ per round, $O(\log T)$ rounds.

For Tier 1, $T = O(n \cdot \ell)$ and $C = O(n)$, giving $O(n^2 \ell \log(n\ell))$ total. For Tier 2, $T = O(n^2 \cdot \ell)$ and $C = O(n^2)$, giving $O(n^4 \ell \log(n^2 \ell))$ total.

L.7 Empirical Validation

Our empirical data (Table 7) shows $t_{\text{prove}} \propto n^{2.06}$ ($R^2 = 0.988$), consistent with Tier 1 quadratic scaling for fixed-length inputs. Verification time scales as $n^{0.37}$ ($R^2 = 0.932$), confirming the well-known STARK property that verification is quasi-logarithmic in circuit size. Proof size scales as $n^{0.49}$ ($R^2 = 0.994$), growing slower than $\sqrt{n}$, consistent with the polylogarithmic proof size of FRI-based protocols.

Proposition L.10 (Boundedness)**.** *For the production-scale corpus (630 BPE tokens, sequences up to 100 tokens), all intermediate WFA weights remain bounded by 2^{40}, well within the comparison gadget domain $[0, 2^{62})$. This was verified empirically across all 10,000 triple checks.*

The practical implications: at $n = 1{,}024$ states (sufficient for most NLP metric WFA), batch proving takes ∼27 s with sub-3 ms verification. At $n = 2{,}048$ states, proving takes ∼113 s—expensive but feasible for high-stakes evaluation where verifiability justifies the computational cost.

The sub-linear proof size scaling ($\sim\sqrt{n}$) means that even large circuits produce proofs small enough for on-chain verification or bandwidth- constrained settings.